

Technical Report

Learning in Tandem (LiT): Women's Collectives from MENA & South Asia

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Date: 25 November 2024

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Funded by

CGIAR GENDER Platform

Executive Summary

The project, “Learning in Tandem: Women's Collectives from MENA & South Asia” (hereafter referred to as LiT project), initiated in January 2024 and concluding in November 2024, investigates the transformative role of women's collectives in addressing gender inequities and fostering climate-resilient agricultural practices in South Asia and the MENA region. Cross-regional learnings are expected from the evolution and rich experience of women collectives in India. Funded by the CGIAR GENDER Platform with additional contributions from other CIMMYT projects (e.g., the “Climate Resilient Agriculture in Bihar” project, by the Borlaug Institute for South Asia, Samastipur), the initiative spans India and Morocco. Although Egypt was initially planned to be included, it was avoided due to administrative constraints that limited the fieldwork. The project emphasizes the feminization of agriculture, focusing on Self-Help Groups (SHGs) as vehicles for empowering women, enhancing livelihoods, and promoting sustainable farming. The project objectives were to:

- Examine institutional factors influencing the effective functioning of rural women's collectives
- Identify micro-level drivers of farmer participation in SHGs across India and Morocco
- Assess the role of SHGs in adopting climate-resilient agricultural practices
- Recommend institutional support mechanisms for inclusive, resilient SHGs

Fieldwork for this study was conducted in three key regions: Bihar and Kerala in India, and Khemisset Province in Morocco, each offering unique socio-economic and agricultural contexts. A mixed-methods approach was adopted, integrating focus group discussions (FGDs), household surveys, and secondary data reviews to ensure a comprehensive understanding of women's collectives and their impacts. In Bihar, the study involved surveys of 1,200 households, with 50% of respondents being women, complemented by stakeholder analyses to understand broader institutional dynamics. It was done as part of the CRA project. However, due to administrative constraints, the FGDs could not be conducted in the state, among the collective participants. In Kerala, 52 focus group discussions and 360 household surveys were conducted to capture the experiences and challenges of women in Self-Help Groups (SHGs). In Morocco, the research encompassed 30 focus group discussions and a survey of 250 individuals, focusing on women's cooperatives engaged in diverse agricultural and entrepreneurial activities. This methodology provided robust insights into the functioning and outcomes of women's collectives across varied cultural and geographic settings.

The findings of the study reveal significant insights into the transformative role of women's collectives across diverse socio-economic and agricultural contexts. Economic empowerment emerged as a critical outcome, with SHGs in all regions providing women access to credit, enabling investments in farming, micro-enterprises, and value-added activities. In Kerala, Kudumbashree SHGs focused on collective farming and microfinance to improve income stability, while Moroccan cooperatives excelled in niche markets like argan oil and medicinal plants, creating opportunities for both economic and cultural preservation. In Bihar, SHG participation was linked to increased financial independence and livelihood diversification, particularly among marginalized groups.

Social empowerment was another major outcome, as SHG participation enhanced women's leadership, decision-making, and community engagement. In Kerala, the structured framework of Kudumbashree facilitated grassroots governance and encouraged women to take on leadership roles within their communities. Similarly, in Morocco, grassroots leadership within cooperatives helped women navigate restrictive cultural norms, though limited institutional support posed challenges. In Bihar, women in SHGs gained significant decision-making power within households and agricultural activities, marking a shift in traditional gender roles.

Climate resilience was a recurring theme, with SHGs in all regions promoting sustainable agricultural practices. In Kerala, women adopted crop diversification, organic farming, and indigenous knowledge to adapt to climate variability. Moroccan cooperatives emphasized eco-friendly pest management and water-efficient farming, showcasing innovation in arid and semi-arid contexts. In Bihar, SHGs fostered climate resilience by encouraging women to use shared resources and technologies to enhance productivity and sustainability.

Despite these achievements, the findings also highlighted persistent challenges, such as limited market access, policy delays, and socio-cultural barriers. In Kerala, bureaucratic inefficiencies and market volatility hindered SHG progress, while Moroccan SHGs faced resource constraints and limited infrastructure. In Bihar, caste-based disparities influenced SHG participation and empowerment, with marginalized groups benefiting less from collective activities.

The LiT project successfully highlighted the transformative potential of SHGs in empowering women and improving agricultural sustainability. The findings underscore the need for tailored support systems to address regional challenges and ensure the scalability of SHG initiatives. The report emphasizes several key recommendations to enhance the impact and sustainability of women's collectives in rural development. First, strengthening institutional partnerships is essential to provide consistent support and resources for women's groups, ensuring their long-term viability. Second, targeted training programs should be developed to equip members with the skills needed to navigate market and climate challenges effectively, fostering resilience and economic growth. Third, the promotion of inclusive policy frameworks that integrate gender equity with agricultural innovation is crucial to address systemic barriers and unlock the full potential of women's collectives. By advancing these recommendations, the report contributes to the global discourse on gender equity, rural development, and climate resilience, offering actionable pathways to leverage women's collectives in achieving Sustainable Development Goals.

Chapter 1

Introduction

The Learning in Tandem (LiT) project, running from January 2024 to November 2024, is funded with US\$ 160,000 from the CGIAR GENDER Platform, with the remaining funds sourced from other CGIAR projects. The project focuses on examining the role of women's collectives in fostering gender equity and promoting climate-resilient agricultural practices across Morocco, Egypt, and India. With the growing feminisation of agriculture, Self-Help Groups (SHGs) have become vital in empowering women by enhancing their participation in agricultural activities and decision-making. The report is based on a comprehensive review of literature and field surveys conducted in Bihar and Kerala States of India, as well as in Morocco. This initiative is part of a wide collaboration involving IWMI, CIMMYT, the Australian National University, Kerala Agricultural University, the Borlaug Institute for South Asia, and contributions from consultants and interns.

Objectives

1. Examine the institutional factors contributing to the effective functioning of rural women's collectives in India and the MENA region.
2. Examine the microlevel factors influencing farmer participation in women's collectives in India and Morocco.
3. Assess the role of women's collectives in empowering women and promoting the adoption of climate-resilient agricultural practices.
4. Identify the institutional support mechanisms necessary for ensuring inclusive, productive, and resilient functioning of rural women's collectives in the MENA region and South Asia.

The project utilizes both quantitative and qualitative datasets from Bihar, Kerala, and Morocco. Due to potential delays in securing administrative approval, the survey was not conducted in Egypt. In Bihar, we interviewed 1,200 male and 1,200 female farmers, with 25% of our random sample comprising women group participants. In Kerala and Morocco, focus group discussions and surveys were conducted. Kerala saw 52 discussions and 360 household surveys among participants, while in Morocco, 30 discussions were conducted. In Bihar, we also conducted 450 household surveys, with 60% of participants being members of women's groups.

As part of the project, we organized a conclave titled JWALA ("Ignite the Fire Within") among women participants at the College of Agriculture, Vellayani, Trivandrum, under the umbrella of the Paula Kantor Hub for Gender and Development, CIMMYT. According to their report, the College of Agriculture, in collaboration with the Paula Kantor Hub, hosted the two-day symposium on April 16-17, 2024. The event, organized by students, aimed to empower them to undertake scientific initiatives that can positively impact society. The main objective was to create a platform for women group members, academicians, policymakers, and students to share knowledge and experiences, helping to formulate better strategies and outcomes for the movement. Additionally, we trained 12 BSc Agriculture students as part of the project on "Basic Socioeconomic Research Methods, with a Gender Focus."

Chapter 2

Nature, Functioning and Evolution of SHGs

"Happily enough, competition is not the rule either in the animal world or in mankind. It is limited among animals to exceptional periods, and natural selection finds better fields for its activity. Better conditions are created by the elimination of competition by means of mutual aid and mutual Support." (Kropotkin 1902).

SHGs have roots in the broader concept of mutual aid and collective action, which dates back to ancient societies. Communities across Africa, Asia, and Europe practiced collective resource pooling, primarily to manage risks associated with agriculture, illness, and economic hardship (Ardener and Burman 1995). In China, for instance, early forms of informal collective savings, particularly in the form of *hui* (association) or *shè* (organization), can be traced back centuries (Morio 2018). These associations have long provided a way for individuals to pool resources for various purposes such as investment, risk management, and social welfare. These rotating savings and credit associations played a significant role in financing both small enterprises and personal consumption, especially in times of economic need. Similarly, such rotating savings and credit associations have been widespread across many parts of Africa, known by various names like *Esusu* in Nigeria, *Tontines* in Cameroon and Senegal, and *Susu* in Ghana, providing essential financial services in the absence of formal banking systems (Bouman 1995).

In developing economies, the transition of women's collectives into formalized Self-Help Groups (SHGs) was significantly influenced by the growth of microfinance in the 1970s and 1980s, particularly in countries like Bangladesh (Khandker 2005). This period witnessed the establishment of institutionalized self-help groups to improve economic access for the poor. The development of group-based microfinance initiatives catalyzed this formalization, providing women with access to credit through structured, group-based lending models (Kumar et al. 2021). The group lending model often used joint liability, where all members were collectively responsible if any individual defaulted (Armendáriz de Aghion and Morduch 2000). In Bangladesh, the Grameen Bank, established by Muhammad Yunus in the 1970s, became one of the most prominent examples of group-based microfinance, where women-led groups were formed to access small loans and build collective financial security (Yunus 1999). In Africa, SHGs have been widely promoted as tools for poverty reduction, women's empowerment, and social cohesion. In Kenya, for instance, SHGs became part of the rural development landscape, with a focus on collective savings and economic activities (Gugerty 2007). The SHG model also expanded to Southeast Asia, where the concept was adapted to address local issues like agricultural productivity, climate resilience, and women's livelihoods (Seibel and Torres 1999).

The evolution of microfinance-focused collectives into formalized SHGs allowed rural women to overcome the barriers traditionally faced in accessing financial services. By providing structured, group-based lending and saving models, these SHGs offered an accessible platform for women in informal economies to secure financial resources, fostering economic resilience and community-wide benefits over time. Through regular group interactions, SHGs also facilitated skill-building, social networking, and knowledge-sharing, reinforcing both financial inclusion and empowerment. International organizations such as the United Nations, World Bank, and International Fund for Agricultural Development (IFAD) have supported SHG programs, recognizing their role in advancing women's rights, reducing poverty, and achieving Sustainable Development Goals (SDGs). South-South Cooperation has facilitated the transfer of SHG

models from countries like India to regions in Africa and the Middle East, where the SHG framework is being adapted to local contexts.

2.1. Women Collectives and Movements in MENA region

The MENA region exhibits vast economic diversity, with per capita GDP ranging significantly—from as low as \$1,400 in Yemen to over \$20,000 in oil-rich Gulf states like Qatar, Kuwait, and the UAE (Waha et al. 2017). This economic disparity influences adaptive capacity to climate risks, as wealthier Gulf nations are better equipped to handle these challenges than others like Morocco, Egypt, and Yemen. Women are integral to the MENA's agricultural sector, yet they face significant structural challenges, including limited access to resources, restrictive social norms, and exclusion from decision-making (Abdelali-Martini 2011). In this sub-section, we examine the role of women's collectives and SHGs in the MENA region, their impact on agricultural productivity, climate resilience, and food security, and the transformative potential these collectives hold for advancing gender equality. Since there is limited evidence on the role and functioning of women collectives in the MENA region, we resort to grey literature, including reports and internet documents.

The international organizations recognize the importance of addressing agriculture, food, and water security challenges in MENA by empowering women, who, despite playing a crucial role, remain disproportionately affected by these issues due to limited access to resources and decision-making (Grossman 2023; Salehi-Isfahani et al. 2014). Gender-based disparities in access to resources like land, finance, and agricultural inputs create obstacles that prevent women from fully benefiting from their agricultural labor. These barriers are further compounded by cultural and social norms that limit women's participation in public and economic life, relegating them to lower-paying, labor-intensive roles in agriculture. Climate change, which intensifies droughts, water scarcity, and soil degradation, disproportionately affects women in agriculture (Baruah and Najjar 2022). Against this context, women's collectives and SHGs have emerged, providing rural women with the resources, knowledge, and support needed to improve their livelihoods and resilience.

Women's collectives in the MENA region began as grassroots initiatives focused on creating community solidarity and sharing resources. These collectives later formalized into SHGs, drawing inspiration from successful SHG models in Asia and Africa (IOM 2024). In countries like Tunisia, Algeria, Jordan, and Egypt, these groups began receiving support from NGOs and international organizations such as the Food and Agriculture Organization (FAO) and International Fund for Agricultural Development (IFAD). In Tunisia and Morocco, for example, women's collectives focused on agricultural processing and marketing have been instrumental in helping women gain access to local and international markets (ILO 2023; Tangi et al. 2024). These cooperatives provide training in sustainable agriculture, financial management, and marketing, empowering women with skills that allow them to improve their agricultural productivity and income levels.

Social Empowerment and Gender Equality. SHGs have transformative effects on social structures; membership in SHGs provides women with a support network that fosters solidarity, self-confidence, and leadership skills. Literature on the effects of SHG on women empowerment in MENA regions has been sparse. However, reports are indicating that SHG participation has been associated with increased decision-making power within households, as women gain economic independence and contribute to household incomes. In Lebanon, for instance, initiatives such as the "Arab Women Speak Out" program have empowered women by enhancing

their self-confidence and decision-making abilities, leading to greater involvement in household and community affairs (Balsom 2017). Moreover, SHGs often provide a platform for women to engage in community decision-making and political awareness. In Jordan, for instance, Takatoat (a feminist collective established in 2020), conducts feminist workshops and book clubs. They collaborated with other groups for protests and focused on issues related to gender-based violence, working in partnership with international feminist organizations (Poust 2023).¹ This increased visibility and influence challenge traditional gender norms, demonstrating women's capacity to contribute to public life and governance. The social support offered by collectives also addresses issues such as domestic violence, health, and education. In some MENA countries, organizations (e.g., High Atlas Foundation of Morocco) have focused on collective forming for women's health and well-being, including workshops on reproductive health, legal rights, and vocational skills.

Case Studies of SHG Success in the MENA Region. Several case studies illustrate the positive impact of women's SHGs in the MENA region. In Tunisia, a women's cooperative specializing in olive oil production has enabled its members to secure better prices for their products by connecting directly with buyers, bypassing middlemen. This cooperative also provides training on sustainable olive cultivation and quality control, ensuring that the women produce high-quality, organic olive oil that appeals to international markets. In Morocco, women's collectives in rural areas have gained international recognition for their production of argan oil, which is highly valued in the cosmetics industry. These cooperatives have not only improved women's incomes but also helped to preserve argan forests, which are vital for biodiversity and climate resilience. The success of these cooperatives has led to similar initiatives across Morocco, empowering thousands of women and promoting environmental sustainability. In Algeria, SHGs focusing on food processing have provided women with the skills to produce and market traditional products, such as couscous and preserves. By developing these microenterprises, women gain economic independence and establish a source of stable income, which is especially valuable in rural areas with limited employment opportunities.

Climate Resilience and Sustainable Agriculture in MENA. Climate resilience is a pressing concern in the MENA region, where extreme weather events and water scarcity threaten agricultural productivity. The World Bank highlights that MENA is one of the most vulnerable regions to climate change, facing challenges such as high temperatures, desertification, and water scarcity, all of which adversely affect agriculture (El Bari et al. 2024). Additionally, soils across MENA are experiencing rapid degradation, erosion, and reduced fertility, further impacting agricultural yields (Waha et al. 2017). Women's SHGs and cooperatives can play a crucial role in climate adaptation by promoting sustainable farming practices. For example, in Tunisia, the GDA Sidi Amor cooperative empowers rural women through sustainable cheese-making practices, providing them with economic opportunities while promoting environmental conservation and sustainable agriculture (ILO 2019). In Morocco, SHGs have been involved in sustainable practices, such as organic farming and eco-friendly pest management, which contribute to both climate resilience and biodiversity conservation (Al-Zu'bi and Krishna 2024). Through SHGs, women gain access to climate-related information and agricultural innovations that allow them to adapt to environmental changes more effectively. In some cases, women's collectives have

¹ Another example is *Tadamun*, an organization founded by activist Asma Khader, which focuses on women's rights, offering legal services, educational programs, and hosting seminars. It plays a significant role in Jordan's women's rights movement by forming alliances and engaging in international partnerships (https://sigi-jordan.org/en/page/our_history).

even influenced local agricultural policies, advocating for support and subsidies for sustainable practices. By empowering women with the knowledge and resources needed to implement these practices, SHGs contribute not only to the resilience of individual farms but also to broader community resilience against climate shocks.

Challenges and Future Prospects. While women's collectives and SHGs have made notable progress in empowering women across the MENA region, they continue to face formidable challenges. Cultural norms and social resistance often restrict women's participation in both economic and public life. For instance, in rural Egypt, societal expectations often confine women to household roles, limiting their engagement in income-generating activities through SHGs. Additionally, in countries like Morocco, restrictive laws hinder women's rights to own land or access credit, stifling the growth and impact of these groups. This lack of land ownership, for example, prevents many women from farming independently or securing loans, which are typically tied to land as collateral.

Another major obstacle to the sustainability of SHGs is their heavy reliance on external funding from NGOs and international organizations. In Lebanon, for instance, SHGs that focus on vocational training and economic resilience depend on foreign aid and grants. However, fluctuations in funding often disrupt the continuity of these programs, making long-term planning difficult. Without consistent support and capacity-building initiatives, SHGs struggle to maintain their impact over the long term.

Strengthening partnerships with local governments is essential for SHG longevity. For example, in Tunisia, government-backed cooperatives in agriculture have shown more stability, as these partnerships enable collectives to access state resources and infrastructure support. Encouraging community ownership also proves vital; when local communities actively participate in SHG activities, they are more likely to contribute resources and ensure continuity. Developing these partnerships and fostering local involvement are crucial steps toward building a sustainable foundation for SHGs in the MENA region, enabling them to continue their transformative work without being solely dependent on external aid

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2.2. Women's Collectives in India

Financial inclusion has emerged as a powerful tool to address gender disparities and discrimination that persist in rural India, contributing to the marginalization of rural women. Indicators of this marginalization include low literacy rates, limited purchasing power, poor health, restricted access to the formal economy, and limited empowerment within both the family and broader society (Lahiri-Dutt & Samanta, 2006) (Lahiri-Dutt & Samanta, 2006). Self-Help Groups (SHGs) have emerged as a transformative force in India's development landscape, especially for marginalized communities, by promoting economic empowerment, social equity, and financial inclusion (Maity, 2023). Originating in the 1980s and inspired by the success of SHG model under financial inclusion pioneered by Professor Muhammad Yunus using the concept of Grameen Bank in 1983 in Bangladesh, SHGs in India since have flourished. They now have grown into a vast network that connects millions of poor families to formal financial services, fostering a culture of savings, credit access, and group solidarity (Garikipati, 2008). The evolution of SHGs in India offers valuable insights into the role of collective empowerment, institutional support, and community-driven development (Hoffmann et al., 2021; Swain & Varghese, 2013). *Early Beginnings: The Role of MYRADA (NGO) and NABARD* The earliest SHG in India was started in 1972

with the Self-Employed Women's Association (SEWA) in Ahmedabad working with the poor and self-employed women in the informal sector, in terms of getting work, income, and food security. The NGO PRADAN (Professional Agency for Development Action) also created in 1983 and promoted similar SHGs. Similarly, in Southern part of India, 'SADHAN', 'DHAN' foundation and 'ASA' worked to promote SHGs (Singh & Majumdar, 2023).

While the formal concept of SHGs in India first took shape in 1985, spearheaded by the NGO MYRADA. This organization played a crucial role in organizing small groups of marginalized women into collective entities following the collapse of large agricultural cooperatives. By 1990 there were a number of organizations in India, which mobilized the poor around groups which had 15 to 20 women members that met once a month to conduct savings and internal credit activities (Kumar et al., 2021). Members pool their savings, which form a collective fund which is used as a loan to individual members (Garikipati, 2008). Since established in 1985, they have spread rapidly throughout India (Fernandez, 2006; Kumar et al., 2021).

MYRADA recognized that financial resources were not the only requirement for marginalized women to achieve economic stability; they also needed social empowerment and a sense of unity. By forming groups based on mutual trust and affinity, women pooled their resources, developing a shared fund from which members could borrow for their personal needs or entrepreneurial activities (Garikipati, 2012). This grassroots approach allowed women to circumvent traditional financial barriers, fostering both economic and social empowerment (Banerjee et al., 2015; Brody et al., 2015; Jakimow & Kilby, 2006; Kumar et al., 2021; Raghunathan et al., 2019).

In 1987, India's National Bank for Agriculture and Rural Development (NABARD) became an influential supporter of the SHG movement. Recognizing the potential of SHGs to reach the unbanked population, NABARD funded MYRADA with a grant of Rs. 1 million. This financial backing helped MYRADA to scale its SHG model, with a focus on capacity building, savings mobilization, and basic financial literacy. NABARD's support was instrumental in formalizing SHGs as a viable component within India's financial system. By creating a structure around SHG activities, NABARD began building a bridge between informal community groups and formal banking systems, laying the foundation for SHG integration into mainstream financial services.

The SHG-Bank Linkage Programme of 1992: The Institutional innovation

In 1992, NABARD launched the SHG-Bank Linkage Programme, a groundbreaking initiative that connected SHGs directly with formal financial institutions. Under this program, banks could lend to SHGs without requiring collateral, as the collective responsibility of group members reduced the risks associated with lending to the poor (Besley & Coate, 1995; Maity, 2023). The SHG-Bank Linkage Programme was designed to address two primary barriers that low-income families faced in accessing credit: the lack of collateral and the bureaucratic processes involved in securing loans from formal financial institutions. Through this program, SHGs became conduits for financial inclusion, where members could easily access credit for small-scale businesses, agricultural activities, and personal needs (Deepika & Sigi, 2014; Marr & Tubaro, 2011; Panda & Atibudhi, 2011).

The success of this model quickly became evident, as it provided banks with low default rates and high repayment performance, supported by the collective accountability within SHGs (Datta, 2015). Over time, the program served as a template for other countries like Afghanistan and Sri Lanka seeking to implement microfinance strategies and promote financial inclusion at the

grassroots level (Singh & Majumdar, 2023). By eliminating heavy bureaucratic hurdles, the SHG-Bank Linkage Programme allowed millions of families to transition from subsistence to sustainable livelihoods through accessible financial services.

Scaling Up in the Late 1990s and Early 2000s

The late 1990s saw a rapid expansion of the SHG movement, primarily due to the success of the SHG-Bank Linkage Programme and increased support from both governmental and non-governmental organizations. Between 1999 and 2005, the SHG model gained momentum across various Indian states, adapting to the unique needs of rural communities. By 2005, the SHG movement had grown exponentially, with over 1.6 million SHGs established across India, reaching approximately 24 million families. Later it led to the formation of country wide National Rural Livelihood Mission (NRLM) in 2011. Currently, the program had mobilized nearly 100 million women in accordance with 9.04 million SHGs and it has become the world's largest microfinance project (Dey, 2018).

The success of SHGs during this period is largely attributed to the robust support from state governments and NGOs that collaborated to promote SHGs as tools for poverty alleviation (Maity, 2023). Women's Empowerment Project and the Swarnajayanti Grameen Swarozgar Yojana (SGSY) integrated SHGs into broader poverty reduction efforts. Through these initiatives, SHGs not only provided financial resources but also empowered women socially and politically, granting them a voice in community affairs and local governance (Desai & Joshi, 2014; Kumar et al., 2021).

Social Empowerment and Gender Equality

Beyond economic empowerment, SHGs became powerful platforms for social change, particularly for rural women (Desai & Joshi, 2014; Kumar et al., 2019). In traditionally male-dominated rural areas, SHGs provided women with a space to make financial decisions, develop leadership skills, and participate in local governance. Membership in an SHG often marked the first time that women handled financial transactions independently, making SHGs crucial for shifting gender norms and promoting gender equity (Kabeer, 2011; Ramachandar & Pelto, 2009; Sahu & Singh, 2012). The influence of SHGs extended beyond finance; these groups became a mechanism for addressing social issues such as education, healthcare, community development, tackling social issues ranging from dowry and domestic violence to gender and caste based discrimination (Chen et al., 2007; Gugerty et al., 2019; Nichols, 2021). A recent study explored whether SHGs can serve as effective platforms for delivering information and disseminating new agricultural technologies to women farmers in India (Raghunathan et al., 2019). This approach aligns with growing recognition of the importance of local knowledge, social networks, and participatory training—elements often neglected in traditional extension services—as critical factors in technology adoption (Chambers & Pretty, 1993; Conley & Christopher, 2001; K. Munshi, 2004).

Through regular meetings and group discussions, SHG members could share knowledge and access resources on a wide range of issues, creating an ecosystem where social and economic goals could be pursued simultaneously. Women in SHGs began advocating for their rights, participating in local decision-making processes, and influencing policies at the grassroots level, all of which contributed to an enhanced social status for women (S. Munshi & Bhardwaj, 2021). *Global Recognition and Support from IFAD & World Bank*

The success of SHGs in India attracted global attention, and international organizations such as the International Fund for Agricultural Development (IFAD) played a pivotal role in further expanding SHGs to marginalized and remote areas. IFAD supported SHG-based projects as part of its commitment to rural poverty alleviation, recognizing that SHGs were effective in reaching under-served populations. IFAD-backed initiatives integrated SHG models into state-sponsored programs and international efforts across Asia, fostering knowledge-sharing and best practices (Fernandez, 2006).

Adding to this momentum, the World Bank made a landmark \$750 million investment in India's National Rural Livelihoods Mission (NRLM) that seeks to transform the lives of poor rural women by organizing them into self-help groups and building resilient community institutions. Together, these international alliances have accelerated the SHG movement, enabling rural women to gain financial independence and drive transformative change in their communities¹. Through collaborations with international organizations, SHGs in India began influencing similar models in other developing nations, becoming part of a global movement for microfinance and women's empowerment. This exchange of ideas reinforced the adaptability of the SHG model, proving that it could be tailored to different cultural contexts while maintaining its core principles of collective savings and financial accessibility.

Challenges and Sustainability Concerns

Despite the successes of SHGs, challenges remain in sustaining and scaling these groups effectively. Some SHGs have struggled with issues related to institutional capacity-building, governance, and over-reliance on subsidies in government programs. While government support has been critical in promoting SHGs, excessive reliance on external funding has, in some cases, weakened the autonomy and cohesion of SHGs. These challenges highlight the need for enhanced training programs and capacity-building efforts to ensure that SHGs remain resilient and sustainable.

Moreover, the scalability of SHGs is often constrained by disparities in local governance and varying levels of institutional support. To address these issues, experts suggest that SHG members be given ongoing training in areas such as financial literacy, governance, and conflict resolution to enhance their resilience and cohesion.

Moreover, social norms in conservative communities present additional challenges. De Hoop et al., (2014) observed negative impacts on the well-being of SHG women in conservative areas, attributing this to social sanctions against women displaying autonomous behaviors. In deeply patriarchal settings, these women face internal struggles and external pressure, as their independence often clashes with societal expectations (Ahmed et al., 2001; Kabeer et al., 2019). These social constraints can threaten SHG sustainability, and such adverse effects are often underreported, as research typically does not focus on potential negative outcomes.

Group heterogeneity also poses a challenge. (Lahiri-Dutt & Samanta, 2006) in her study of SHGs in West Bengal, identified lack of coordination, poor communication, and transparency issues in failed groups. Leadership struggles, often compounded by interference from family members, and divisions based on class, caste, and religion further exacerbated group instability. These factors highlight the need for strong, inclusive leadership and transparent processes to build lasting SHG structures.

Chapter 3

Field Study Methodology

3.1. Morocco

Study Area

To finalize the selection of the research area, discussions were held with officials from the Ministry of Agriculture at the central level, particularly with representatives of the Strategy and Statistics Department (DSS) and the Agricultural Development Agency (ADA). Among the options considered, the province of Khemisset was selected based on its alignment with the research objectives, the significant presence of women's agricultural cooperatives, and the vulnerability of its rainfed agricultural systems to climate change. Due to persistent drought conditions in recent years, farmers (both men and women) and their cooperatives have been adapting their cultivation and post-harvest methods to sustain their family incomes.

Located in the Rabat-Salé-Kénitra region of Morocco, Khemisset province features diverse geographies, including high-altitude plateaus (950–1300 m), sandy plains, the Zaer plateau, and arid zones. Agriculture in this region is predominantly rainfed, with only 5% of its usable agricultural area under irrigation. Key activities include cultivating cereals, legumes, and olive trees, raising livestock, and producing aromatic and medicinal plants. These practices are highly vulnerable to recurring climate challenges, particularly droughts, which have profound impacts on cropping patterns and the resilience of farming communities. Khemisset serves as a critical area for studying the socio-economic and environmental factors affecting rural livelihoods in Morocco's rainfed regions.

According to 2022 data from the Provincial Department of Agriculture in Khemisset, the province has a population of 542,221, of which 51.84% live in rural areas. Agriculture is predominantly small-scale, with 60% of farms covering less than 5 hectares. Cooperative activity includes 270 registered cooperatives, approximately 20% of which are led by women. These women-led cooperatives focus on activities such as food processing, animal husbandry, apiculture, and the production of medicinal and aromatic plants. Despite systemic challenges like limited access to resources and markets, these cooperatives play a crucial role in local economies.

Focus Group Discussions

The research adopted a participatory approach to survey agricultural cooperatives in Khemisset, with a focus on empowering rural women. Preparatory steps included a field survey and a review of the Provincial Department of Agriculture's monograph. Secondary data were collected on village distribution, women's cooperatives, and production systems. To refine the fieldwork, an information session was held with the Director of Agriculture and his staff to review research objectives and finalize the selection of villages and cooperatives.

The cooperatives were chosen based on three criteria:

1. Majority of members being women.
2. Diversified income-generating activities within the sample.
3. Representation across the agro-ecological zones of Khemisset.

Discussions with the Director of Agriculture reinforced the relevance of the study, especially given the priority placed on empowering rural women in the context of climate resilience by public authorities.

Sampling Methodology

The study selected 30 women's cooperatives in Khemisset for focus group discussions (FGDs). The selection criteria were:

1. Active for at least three years.
2. Engaged in activities such as food processing, agriculture, apiculture, animal husbandry, or medicinal and aromatic plants production.

The distribution of cooperatives in the sample is as follows:

- Aromatic and medicinal plants: 7 (23.3%)
- Beekeeping: 7 (23.3%)
- Couscous processing: 7 (23.3%)
- Agriculture: 4 (13.3%)
- Animal products: 3 (10%)
- Eco-tourism: 1 (3.33%)
- Carpet weaving and embroidery: 1 (3.33%)

Each cooperative had an average of 8 women members, with a range of 5 to 20. The total sample included 234 members, 80% of whom were women, with an average age of 44 years. Most women (65.8%) were over 40 years old, while 7.3% were under 25, and 26.9% were aged 25–40. Marital status data revealed that 65% were married, 24.3% single, 7.3% widowed, and 4.3% divorced. Additionally, 78% lived in nuclear families, while others shared housing with in-laws.

Educational attainment was low, with 41% of women being illiterate. The rest had varying levels of education: primary (22%), secondary (14%), high school (10%), and university (13%). Regarding work outside household activities, 50% were self-employed in post-harvest operations, 35% worked in agriculture, 9% in livestock breeding, and 6% in non-agricultural activities. None were employed as hired agricultural labor. Notably, only 2% held diplomas as agricultural technicians, while 28% had training from the Ministry of Agriculture, and 70% had no training. FGDs involved approximately two-thirds of active members from each cooperative, providing insights into group dynamics, governance, and challenges.

Household surveys

In addition to FGDs, a household survey was conducted with 194 women and 56 men from 28 villages (250 individuals). This survey collected data on women's participation, impacts, and perceptions of cooperative activities. Semi-structured interviews explored group formation, operations, governance, and challenges faced by women's cooperatives. Survey questionnaires were originally in English and were translated into French for the enumerators. Training sessions were conducted in French and Arabic to facilitate effective communication with respondents. Six graduate enumerators (2 women and 4 men) with extensive field experience and local knowledge were supervised by a female coordinator. The training also tested the questionnaires and clarified complex questions.

Primary data collection began with the selected women's cooperatives and was followed by interviews with 250 individuals from the target villages. Each survey took 3–4 hours on average, with some FGDs extending to 5 hours due to members' daily cooperative activities. Surveys were conducted between September 2 and November 3, 2024, accommodating availability challenges that caused delays. Survey responses were coded and entered into Excel under the supervision of the study manager. Data cleaning included post-survey verification with respondents. The complete dataset, including FGDs and household surveys, was translated back into English to meet study requirements.

3.2. Kerala

Guided by the feminist empowerment framework, this study examines the intricate relationships between Self-Help Group (SHG) participation and rural women's empowerment. The research aims to:

1. Explore rural women's aspirations and livelihood strategies shaped by SHG participation.
2. Analyze the constraints and opportunities affecting SHG effectiveness.
3. Investigate the impact of SHG participation on women's economic empowerment and overall well-being.

A mixed-methods research design was adopted, integrating household-level quantitative surveys with qualitative approaches. This methodology enables a detailed understanding of the challenges, opportunities, and impacts of SHG participation, laying a robust foundation for policy recommendations and practical interventions.

Study Area

Located on India's southwestern coast, Kerala spans 38,863 sq. km and is characterized by diverse topography and ecosystems, including coastlands, wetlands, plains, and hills. While historically rooted in agriculture, the state now faces challenges such as labor shortages, marketing inefficiencies, and climate uncertainties. For this study, Malappuram and Thiruvananthapuram districts in Kerala were purposively selected due to their distinct attributes:

- Malappuram: As the district where *Kudumbashree* was first implemented, it is predominantly rural and culturally diverse, making it an ideal location for examining the grassroots impact of *Kudumbashree* (Irshad & Juman, 2015).
- Thiruvananthapuram: The state capital represents an urban-rural nexus, with numerous initiatives providing insights into the broader role and impact of women's collectives in both urban and semi-urban settings.

Focus Group Discussions

To investigate the role of women's collectives in sustainable agriculture and food security, qualitative surveys involving 52 Focus Group Discussions (FGDs) were conducted with *Kudumbashree* groups in both districts. Groups focused solely on microfinance or non-agricultural enterprises (e.g., textile units) were excluded. A sample list of SHGs was generated through consultations with key informants, including:

- *Kudumbashree* State and District Mission personnel,
- *Kudumbashree* Chairpersons,
- Community Development Society (CDS) representatives, and
- SHG members.

The selected SHG units represented a diversity of performance levels and activity types. FGDs were guided by a semi-structured questionnaire to analyze successful and unsuccessful groups, focusing on factors influencing performance, constraints faced, and strategies for marketing, upgrading, and climate adaptation in agriculture and allied enterprises. In-depth interviews with select members further clarified leadership roles and other emerging aspects. Ethical considerations, including voluntary participation and confidentiality, were emphasized during every session. Data from FGDs was coded and analyzed using NVivo software.

- Thiruvananthapuram (23 FGDs): Groups engaged in collective farming (e.g., floriculture and vegetable production), food processing (e.g., *Nutrimix* production), and catering services, including community kitchens.
- Malappuram (29 FGDs): Groups engaged in collective farming (e.g., paddy and fruit cultivation), food processing (e.g., spice mix production), and catering services.

Each FGD involved 4 to 10 participants and lasted 1.5 to 2 hours. Insights from these FGDs informed the selection of Malappuram for the subsequent household survey phase.

Household surveys

Malappuram District in Kerala, a key region for the establishment of *Kudumbashree*, a state-led poverty eradication and women empowerment initiative, was selected for the household survey. The district's active participation in *Kudumbashree* SHG initiatives, combined with its diverse socio-economic profile and robust grassroots infrastructure, provides an ideal context for analyzing the functioning and impacts of SHGs.

This study employed a multi-stage sampling design, using Community Development Blocks (CD Blocks) and Panchayats as the primary sampling frames to select representative SHGs. The process ensured geographical dispersion and diversity across Malappuram District:

1. Selection of CD Blocks: Four CD Blocks—Nilambur, Areekkode, Kalikkavu, and Tirur—were identified as the initial sampling units, ensuring representation of diverse geographies.
2. Selection of Panchayats: Panchayats within the selected CD Blocks were randomly chosen, providing a mix of rural and semi-urban contexts.
3. Selection of SHGs: A purposive sampling approach was employed to select SHGs within the Panchayats, based on their unique characteristics and activities.
4. Selection of Respondents: SHG members directly engaged in specified activities were identified for participation. Approximately 90 SHG members were selected from each of the four CD Blocks, yielding a total sample size of 361 participants.

To capture the heterogeneous dynamics of SHGs, stratified sampling was applied across four activity-based categories:

1. Collective Farming: Sustainable agriculture and collaborative farming practices.
2. Micro-Enterprises: Food processing, value addition, and small-scale entrepreneurship.
3. Community Kitchen Initiatives: Member-owned and member-operated catering services.
4. Microfinance Activities: Financial intermediation, thrift, and credit services.

This stratified approach ensured that the selected SHGs reflected the diverse social, economic, and entrepreneurial contexts of Malappuram District, enhancing the external validity of the study. The research commenced with preliminary discussions with *Kudumbashree* Mission authorities at the state and district levels, providing critical insights into the local context. Fieldwork was conducted from August 28 to September 30, 2024, across 28 Panchayats strategically distributed within the four selected blocks. Data collection employed a digital approach using standardized, pre-tested questionnaires administered by trained enumerators via tablets. The use of *KoboToolbox* software facilitated real-time data validation, reducing errors and ensuring data quality. Each survey took approximately one hour to complete. A total of 361 SHGs were surveyed, including 131 units in collective farming, 114 in food processing, 47 in catering services, and 69 in microfinance.

Supervisory oversight during fieldwork ensured strict adherence to protocols, while enumerators followed standardized procedures to maintain consistency. A rigorous two-stage data quality process was implemented:

1. Initial Review: Raw data was checked for completeness and accuracy, with discrepancies resolved through cross-validation with field notes and follow-up with respondents.
2. Advanced Data Cleaning: Variables were standardized, data was formatted for analytical tools, and the dataset was prepared for long-term preservation using established

ontologies and metadata standards. This meticulous process guaranteed data accuracy, reliability, and reproducibility.

The data analysis integrated quantitative and qualitative techniques to provide a comprehensive understanding of the findings:

Quantitative Analysis

- Descriptive statistics and exploratory data analysis were conducted using *Excel*.
- Outputs included demographic profiles, SHG outcomes, trends, and correlations.
- Results were visualized through tables, charts, and figures for clarity and interpretability.

Qualitative Analysis

- Data from in-depth interviews was coded and analyzed using *NVivo* software.
- Themes and patterns, such as economic empowerment, social cohesion, and challenges, were identified.

Quantitative and qualitative findings were triangulated to provide a nuanced understanding of SHG dynamics and facilitated the identification of meaningful relationships and actionable insights. The findings informed conclusive recommendations for policy and practice.

This study adhered to rigorous ethical standards. Participants' rights, dignity, and well-being were prioritized throughout the research process, with informed consent obtained after explaining the study objectives and procedures, participant data anonymized and securely stored to maintain confidentiality, fieldwork conducted with cultural sensitivity to local norms and values, and measures implemented to minimize biases, including enumerator, contextual, sampling, social desirability, and researcher biases.

3.3. Bihar

We didn't get permission from Jeevika organization to conduct focus group discussions among the members. However, we have done a stakeholder analysis and used the primary data collected from 2400 farmers from 1200 households (50% women respondents). Bihar faces significant socio-economic challenges, with high poverty rates and low development indicators, particularly affecting Scheduled Castes (SC) and Scheduled Tribes (ST). In response, SHGs have been promoted as a poverty alleviation strategy through government-led and internationally supported programs.

The study employed a multi-stage random and purposive sampling approach. Ten districts from Bihar's four agro-climatic zones were selected, and eight villages were chosen from each district. Within each village, a random sample of ten SHG households was selected, supplemented by an additional five households, including lead farmers and female SHG members, to ensure representation across SHG and non-SHG households. The final sample comprised 1,200 households and 2,400 individual respondents. A structured questionnaire facilitated data collection, gathering information on household characteristics, socio-economic factors, SHG dynamics, and participation metrics. Trained local enumerators conducted interviews, with male and female enumerators assigned to male and female household heads, respectively.

In light of the influence of caste structure and socio-economic factors, the study integrated variables such as caste, wealth, education, and landholding into the analysis. A probit regression model was applied to identify factors that affect the likelihood of SHG participation, with interaction terms to examine how caste interacts with wealth, age, and education. The probit

model was also utilized as the first stage in a Heckman two-step procedure, which examined SHG participation likelihood and intensity, focusing on respondents' engagement in farming and leadership roles within SHGs. This approach provides a comprehensive understanding of the role of SHGs in addressing caste-based disparities and broader socio-economic challenges faced by rural women in Bihar.

In the second part of the Bihar Story we focus on examining the effects of SHG participation on women's empowerment and decision-making across various domains of household and agricultural activities. The study uses the Women's Empowerment in Agriculture Index (WEAI) as a primary tool, measuring empowerment across multiple areas such as production decisions, resource control, and time allocation. To analyze differences in empowerment between SHG participants and non-participants, chi-square tests assess the statistical significance of observed variations in activity participation and decision-making authority. Additionally, regression models evaluate the empowerment score differentials across caste groups, showing how SHG membership affects intra-household equality and income control, thus highlighting caste-based disparities. This methodology provides a nuanced understanding of SHG participation's role in improving decision-making power and reducing gender inequities within households.

Chapter 4

Working with SHGs: The Bihar Story

The Self-Help Group (SHG) movement in Bihar, especially under the Jeevika program, has become a powerful vehicle for poverty alleviation and women's empowerment. Supported by the World Bank and the Government of Bihar, Jeevika aims to improve socio-economic conditions by mobilizing rural women into SHGs, fostering financial inclusion, and promoting sustainable livelihoods. This initiative has expanded significantly since its inception, building a robust network that integrates financial literacy, skill development, agricultural extension, and health awareness. The program targets marginalized communities, particularly women from lower socio-economic backgrounds, and emphasizes the role of SHGs in enhancing decision-making, community engagement, and economic resilience. Through multi-level governance structures, Jeevika leverages a bottom-up approach to empower women, helping them transition from basic financial support to broader socio-economic roles, including participation in Farmer Producer Organizations and entrepreneurship.

Findings

Table 1 presents a comparative analysis of SHG characteristics across caste groups, highlighting differences in membership dynamics, decision-making authority, and financial contributions. Notably, Other Socially Marginalized Castes (OSMC) exhibit the highest levels of caste homogeneity within groups, with 86% of SHGs in this category comprised of members from the same caste, compared to 65% in SC/ST groups and 56% in Non-Marginalized Categories (NMC). The data indicate that caste-based clustering within SHGs is a significant factor, with OSMC groups showing statistically significant intra-group cohesion ($p < 0.01$). Decision-making within groups also varies across castes; SC/ST members reported slightly higher rates of sole decision-making authority regarding SHG membership at 73%, compared to 68% among OSMC members and 70% in NMC groups. These findings suggest that, while SHGs are designed for collective action, caste influences persist in shaping group composition and individual agency within SHGs.

In terms of financial contributions, monthly savings rates are relatively consistent across groups, with a marginally higher average balance in OSMC groups (\$38.75) compared to SC/ST and NMC groups (\$34.13 and \$31.12, respectively). Loan accessibility shows minimal variation across castes, suggesting equitable access to credit resources within SHGs. However, the presence of a caste-based stratification in group composition may influence collective decision-making and trust within SHGs, impacting the potential benefits derived from these collectives. These findings highlight the need to consider caste as a mediating factor in SHG dynamics, with implications for policies aiming to enhance inclusivity and empowerment in collective rural organizations.

Figure 1 illustrates the distribution of SHG group activities by caste, shedding light on how participation in specific group activities varies among caste categories. The figure reveals that OSMC members participate more frequently across most SHG activities compared to SC/ST and NMC members. Activities such as savings mobilization, loan repayment, and mutual assistance reflect particularly higher engagement among OSMC participants, with statistically significant differences observed ($p < 0.05$ and $p < 0.01$). SC/ST members, while actively engaged, show comparatively lower participation in leadership roles and decision-making activities within the group, indicating possible disparities in empowerment within SHGs. This distribution highlights that, although SHGs aim to provide equal opportunities, caste-based disparities continue to

influence the level and type of engagement members have in collective group activities. The variations suggest that SHGs may offer differentiated levels of empowerment depending on caste composition, raising important considerations for policies seeking to leverage SHGs as vehicles for inclusive rural development.

Table 4.1. Characteristics of SHG groups by caste

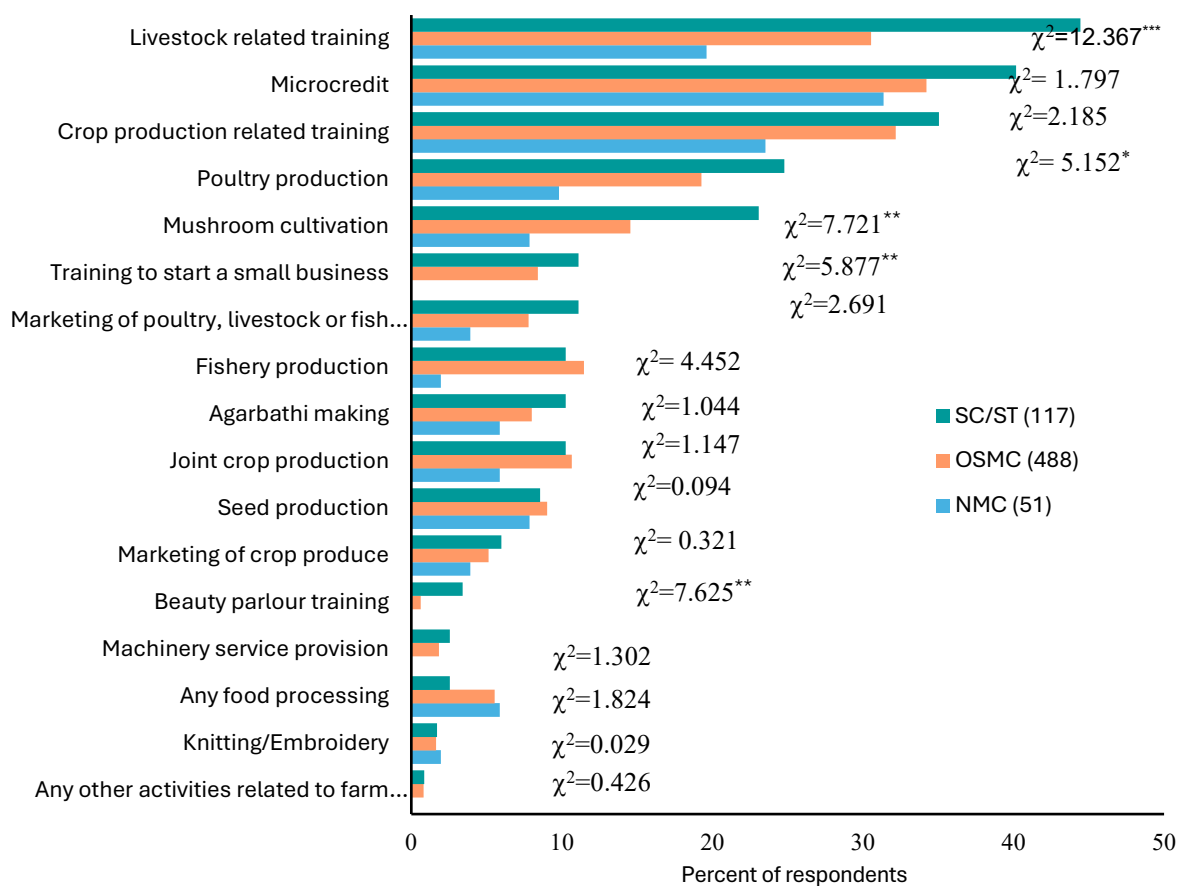
		Caste categories		
		NMC (n=43)	OSMC (n=499)	SC/ST (n=114)
Year of starting of the group in the village (Median)		2017	2017	2017
Number of members in the SHG (number)		12.00 (1.35)	11.61 (2.27)	11.58* (1.34)
Group consists of members from the same caste group only (dummy, 1 yes, 0 otherwise) [#]		0.56	0.86	0.65
Is there a membership fee to join the group? (dummy, 1 if yes; 0 otherwise) ^{##}		0.00	0.05	0.04
Decision of joining the group (dummy)	1 if respondent solely takes decision, 0 otherwise	0.70	0.68	0.73
	1 if spouse takes decision, 0 otherwise	0.30	0.28	0.23
	1 if other household member takes decision, 0 otherwise	0.00	0.03	0.04
Monetary contribution by each individual member to the group (USD\$/month)		0.51 (0.08)	0.50 (0.12)	0.49 (0.07)
Amount of savings balance in the group at present (USD\$)		31.12 (16.69)	38.75* (28.72)	34.13 (20.45)
Family has an outstanding loan from the group (dummy, 1 if yes, 0 otherwise)		0.67	0.64	0.63
Spending of money obtained as a loan (dummy)	1 if respondent solely takes decision, 0 otherwise	0.52	0.44	0.49
	1 if spouse takes decision, 0 otherwise	0.48	0.49	0.49
	1 if other household member takes decision, 0 otherwise	0.00	0.06	0.28
Number of other such groups in the village (number)		10.86 (4.20)	11.16 (6.19)	11.43 (5.80)

Notes: SC stands for scheduled castes, ST for scheduled tribes, OSMC for other socially marginalized castes or OBC in government records, NMC for non-marginalized castes. The inter group difference (with the NMC category as the reference group) is assessed using t mean comparison test for continuous variables. * denotes statistical significance of the difference from the NMC category at the 0.10 level. For categorical variables, a Chi-squared test is used to evaluate significant differences across caste groups. [#]variable 'members from the same caste group' shows a statistically significant difference (Pearson $\chi^2(2) = 12.5519$, $Pr = 0.002$) across caste groups

^{\$}conversion rate of the INR to the US Dollar is 1 INR = 0.012 Dollars (2023).

Source: <https://data.worldbank.org/indicator/PA.NUS.FCRF>

Fig 4.1. Group activities of SHG respondents by caste (2023)



Notes: The vertical axis shows the percentage of SHG respondents to group activities they involved in 2023. The χ^2 statistic for each group activity is provided along with its significance level. ** and *** indicate that the difference across caste groups is statistically significant at the 0.05 and 0.01 levels, respectively.

Table 4.2 presents the marginal effects from a probit regression model identifying determinants of women's participation in SHGs, revealing several key socio-economic and demographic factors. Caste is a significant predictor, with women from OSMC and SC/ST groups showing a higher likelihood of SHG participation compared to those from non-marginalized categories, highlighting the targeted inclusion of socially marginalized groups in SHG programs. Economic status, measured through an asset index, also significantly influences participation, with lower asset scores increasing the likelihood of joining SHGs, suggesting that economically disadvantaged women are more inclined to participate.

Age and education show a complex relationship with SHG membership. Age has a negative linear effect but a positive squared effect, indicating a non-linear relationship where younger and older women are less likely to join than those in middle age. Similarly, while education negatively impacts participation initially, a higher level of education beyond a threshold positively influences membership. The inclusion of interaction terms, such as age and caste or education and caste, further underscores how these demographic variables interplay with caste to shape participation. Overall, these findings suggest that SHG membership is not only influenced by individual socio-economic characteristics but is also moderated by intersectional factors, which must be considered in SHG policy design to improve inclusivity.

These findings are enriched by incorporating insights from Figure 4.2, which presents the predicted marginal effects of interaction terms on SHG membership, adding depth to our understanding of how caste intersects with age, education, economic status, and land ownership in determining SHG participation.

Figure 2a shows that the probability of SHG membership varies significantly by age across caste groups. SC/ST women, for example, demonstrate higher participation rates at younger ages compared to non-marginalized groups, but this likelihood diminishes with age. This trend suggests that SHGs may be especially relevant as empowerment avenues for younger women in marginalized communities, potentially addressing both economic and social vulnerabilities earlier in life. Figure 2b highlights the role of education, where we observe a positive relationship between education and SHG membership across castes, particularly for SC/ST women. For these women, an increase in education substantially raises SHG participation likelihood, underscoring the importance of educational attainment as a catalyst for empowerment within marginalized groups. This finding aligns with literature indicating that education enhances agency, equipping women with skills and confidence to engage in collective organizations.

Figure 2c demonstrates how economic status, measured by asset index, impacts SHG participation across caste lines. For economically disadvantaged groups, especially among SC/ST women, the probability of membership rises as economic status declines, indicating that SHGs attract those with limited economic resources. This pattern suggests that SHGs serve as critical support structures for economically marginalized women, providing access to credit, savings, and potentially, livelihood opportunities, which may otherwise be inaccessible. Lastly, Figure 2d illustrates the effect of land ownership on SHG membership, showing that women from OSMC and SC/ST groups with smaller landholdings are more likely to join SHGs compared to those with larger holdings. This trend reflects the role of SHGs in catering to the needs of land-poor rural women, especially from marginalized communities, who may rely on these groups for financial and social capital that bolster resilience and income stability.

Table 4.2. Determinants of women's participation in SHG: Probit model marginal effects

Variables	Model 1	Model 2	Model 3	Model 4
Household characteristics				
Caste	OSMC		0.177*** (0.043)	0.104** (0.042)
	SC&ST		0.106* (0.057)	-0.016 (0.058)
Asset index score			-0.629** (0.249)	-0.503** (0.250)
Asset index score square			0.408 (0.347)	0.278 (0.341)
Owned land			-0.084*** (0.016)	-0.063*** (0.018)
Other household characteristics	No	No	No	Yes
Respondent characteristics				
Age	-0.005*** (0.001)	-0.005*** (0.001)	0.045*** (0.007)	0.036*** (0.007)
Age square	-4.E-04*** (8.E-05)	-4.E-04*** (8.E-05)	-5.E-04*** (8.E-05)	-4.E-04*** (8.E-05)
Education	-0.017*** (0.003)	-0.017*** (0.003)	0.021** (0.009)	0.017* (0.009)
Education square	-0.002*** (8.E-04)	-0.002*** (8.E-04)	-0.003*** (8.E-04)	-0.002*** (8.E-04)
Interaction (Age*Caste)	No	No	Yes	Yes
Interaction (Education*Caste)	No	No	Yes	Yes
Interaction (Asset index score*Caste)	No	No	Yes	Yes
Interaction (Owned land*Caste)	No	No	Yes	Yes
District dummies		Yes	Yes	Yes
Observations	1200	1200	1200	1200

Notes: Coefficients with standard errors are given in parenthesis. *, **, *** represent significance at 0.10, 0.05 and 0.01 levels, respectively.

Model 4 also serves as the result of the first step (selection equation) in the Heckman two-step procedure.

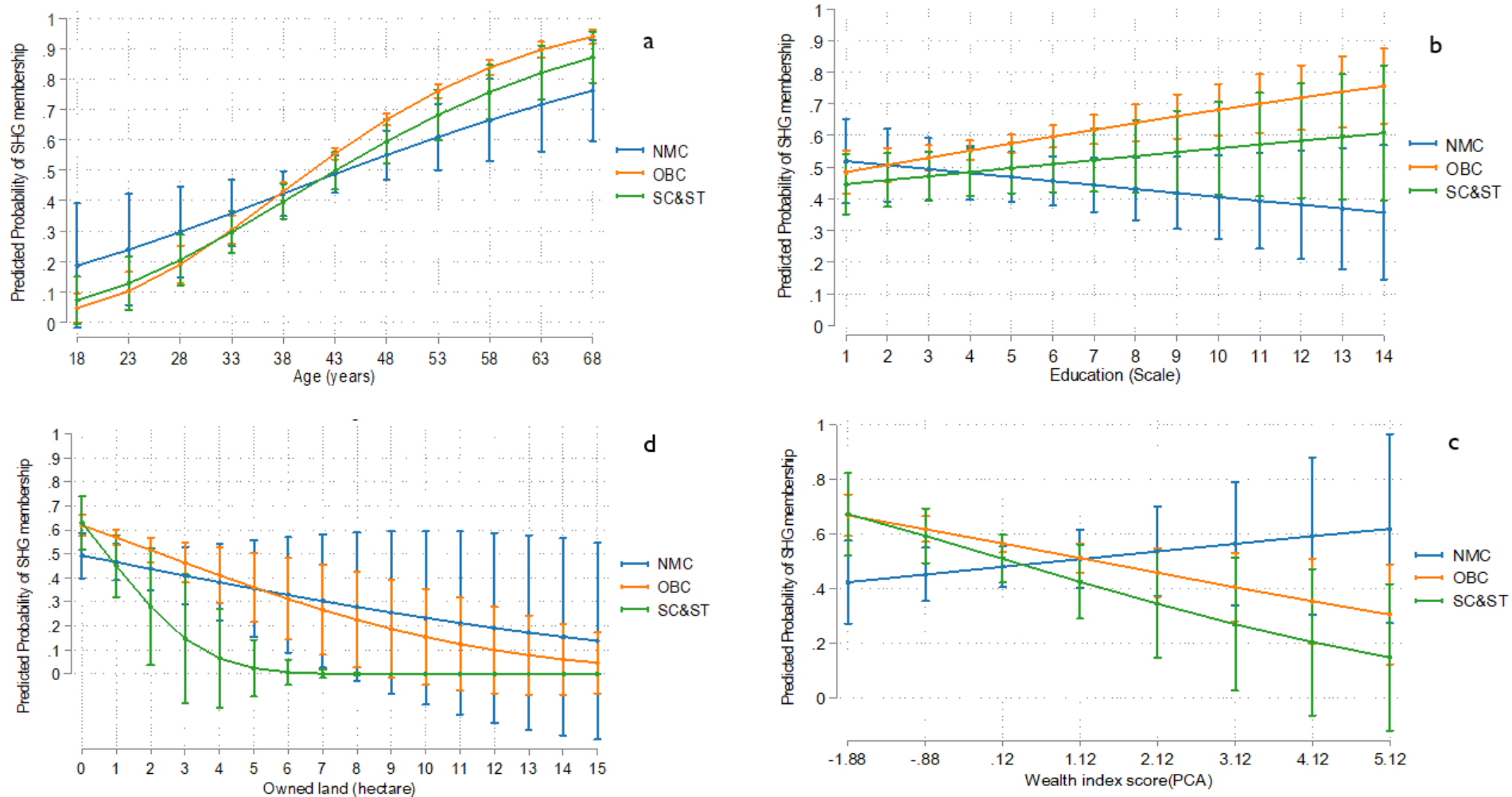


Figure 4.2. Predicted margins of interactions terms: Caste and Age (2a), Education (2b), Economic status of women respondents on SHG membership (2c) and Owned land (2d).

Notes: The figure shows marginal effects calculated at sample mean values and adjusted predictions with 95% confidence intervals. The vertical axis shows the predicted probability of SHG membership. Source: Estimated from regression models (4) in Table 3

Table 4.3 shows decision-making levels across various activities highlights a notable difference between SHG participants and non-participants. SHG participants generally exhibit higher levels of involvement in decision-making, particularly in agricultural activities such as plot selection, crop selection, and access to seeds. For instance, in food crop decisions, only 2.59% of non-participants make decisions independently, compared to 10.29% of SHG participants. This trend is consistent across other areas, such as cash crops, where 7.35% of SHG participants independently choose the crop variety versus only 1.05% of non-participants.

In livestock management and marketing, SHG participants also show increased involvement, with a larger share making independent decisions regarding the sale of milk and animals. Moreover, SHG participants demonstrate greater autonomy in managing major and minor household expenditures, which suggests that SHG membership enhances women's influence over household financial decisions. Overall, these findings underscore the empowerment effect of SHGs in increasing women's active participation and decision-making power across various domains.

The Nearest Neighbor Matching (NNM) estimates from Table 4.4 illustrate the impact of SHG participation on empowerment outcomes across gender and caste lines, revealing significant benefits for women's empowerment and reductions in gender inequality within households. The empowerment score (4DE) for SHG-participating women shows a substantial positive effect, with an NNM estimate of 0.282. This increase indicates that SHG membership enhances women's decision-making power, confidence, and autonomy within the household. For men in SHG-participating households, the empowerment score also rises, albeit modestly (0.036), suggesting that while women's empowerment is more directly impacted, men may also experience indirect benefits in shared decision-making environments. Additionally, the intrahousehold inequality score shows a statistically significant reduction with an NNM estimate of -0.183, signalling that SHG involvement contributes to decreasing gender disparities within the household and elevates women's status relative to men.

The table also provides insights into the effects of SHG participation across different caste groups, further highlighting the differential impact of SHGs based on social stratification. Among women in Other Socially Marginalized Castes (OSMC) and Non-Marginalized Category (NMC) groups, the NNM estimates show significant empowerment gains of 0.226 and 0.228, respectively, underscoring SHGs' role in advancing women's agency across varied social strata. In contrast, the impact on men's empowerment scores varies by caste group, with a minor negative effect observed for NMC men (-0.078), possibly reflecting a rebalancing of decision-making power within these households in favor of women's empowerment.

Table 4.3. Level of decision-making across activities by SHG participants and non-participants

Activities		Participants (n=653)				Non-participants (n=546)				χ^2 test
		Not partici pated	Spouse/ other members	Herself and spouse jointly	Self	Not partici pated	Spouse/ other members	Herself and spouse jointly	Self	
Food crops	Plot selection	43.01	29.03	17.67	10.29	65.25	20.89	11.28	2.59	***
	Crop/variety selection	54.07	22.27	13.36	10.29	71.72	18.11	7.95	2.22	***
	Seed source selection	78.65	6.14	4.92	10.29	90.76	4.07	3.51	1.66	***
	Who access seeds	84.64	3.69	2.61	9.06	94.27	2.22	2.03	1.48	***
	Grain for consumption	13.67	23.20	32.26	30.88	25.69	27.73	26.06	20.52	***
	Grain for marketing	62.67	13.52	14.90	8.91	71.90	16.08	10.17	1.85	***
Cash crops	Plot selection	53.15	25.42	14.08	7.35	71.60	15.97	7.14	1.05	***
	Crop/variety selection	66.18	16.18	10.29	7.35	82.47	9.45	4.83	0.63	***
	Seed source selection	82.77	4.62	5.88	6.72	95.06	2.52	1.05	0.63	***
	Who access seeds	87.82	2.94	3.15	6.09	95.80	2.10	0.84	0.63	***
	Grain for marketing	57.98	18.28	16.18	7.56	70.12	16.18	7.56	1.68	***
Cattle raring	Marketing of animals	66.09	14.94	10.92	8.05	73.73	12.77	10.36	3.13	***
	Sale of milk	46.36	14.18	14.37	25.10	59.04	15.42	11.08	14.46	***
Wage/salary employment		80.28	3.21	2.98	13.53	83.62	4.10	2.05	10.24	
Non-farm activities		78.47	7.18	4.31	10.05	83.54	10.13	1.90	4.43	*
Major expenditure		45.54	25.50	16.83	12.13	51.19	27.98	15.48	5.36	***
Minor expenditure		1.99	14.83	14.68	68.50	5.72	24.17	10.89	59.23	***

Notes: Percentages are conditioned on participation in different activities. Hence, the number of observations varied across activities between SHGs participants and non-participants. Level of participation is given in Figure 1. We have excluded fish farming and poultry farming from our analysis as these activities are not common in the study area. *** and * denotes that the level of involvement in decision-making between SHG participants and non-participants varies across different activities is statistically significant at 0.01, and 0.10 levels, respectively.

Table 4.4. Effect of SHG participation on women's and men's empowerment across caste groups, nearest neighboring estimates

Outcomes	Women				Primary men from the women respondent [#]			
	Overall	ST/SC	OSMC	NMC	Overall	ST/SC	OSMC	NMC
Empowerment score (4DE)	0.282*** (0.022)	0.255* (0.107)	0.226*** (0.025)	0.228* (0.094)	0.036** (0.015)	0.015 (0.098)	0.006 (0.016)	-0.078* (0.036)
Intrahousehold inequality score (Based on the dual-adult household sample, N = 1151)	-0.183*** (0.025)	-0.186 (0.145)	-0.168*** (0.026)	-0.247* (0.093)				

Notes: [#]Either spouse or primary male decision-maker within the household. Standard errors in parenthesis. ***, and * represent significance at 0.01, and 0.10 levels, respectively.

Conclusion

The analysis of SHG participation in Bihar underscores the transformative potential of self-help groups in advancing women's empowerment, especially in regions with entrenched social and economic challenges. Through increased decision-making power in agricultural and household activities, SHG membership has facilitated greater agency among women, fostering both economic independence and enhanced social status within their communities. The study's findings reveal that SHG participants are significantly more involved in household decisions compared to non-participants, highlighting the role of collective action in strengthening women's voices in the domestic sphere. Additionally, the differentiated impacts across caste groups emphasize the need for SHG interventions to address intersectional barriers and ensure inclusivity.

Despite the promising outcomes, the sustainability of SHGs remains a challenge, particularly due to reliance on external support and funding. Strengthening partnerships with local institutions and fostering community ownership will be essential for the long-term success of SHGs. This chapter's insights contribute to the broader discourse on rural development and gender equality, indicating that SHGs can serve as viable instruments for poverty alleviation, inclusive growth, and social transformation, provided that structural and financial challenges are systematically addressed.

Chapter 5

Working with SHGs: The Kerala Story

[Paper title: Interplay of Institutions: How Formal and Informal Structures Shape Women's Collectives for Food Security in Kerala]

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Abstract

The present study examines the interplay of formal and informal institutions in shaping women-led self-help groups (SHGs) for food security in Kerala, India. Kudumbashree, one of India's most successful women's empowerment programs, has enabled SHGs to transcend traditional roles by fostering collective farming, income generation, and technology adoption. Insights from 52 focus group discussions in Trivandrum and Malappuram, analyzed using NVivo, highlight the program's achievements in building resilient community-driven initiatives while identifying areas for further enhancement. The study shows that formal institutions provide robust support through structured frameworks, capacity-building, and linkages to government programs like Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA). Informal networks foster resilience by promoting knowledge-sharing and community cohesion. These interactions have helped SHGs navigate climatic and market constraints, expand market access, and integrate digital tools to address emerging challenges. However, administrative delays, inconsistent policy implementation, and socio-political biases hinder equitable participation and limit innovation adoption, particularly in advanced technologies like precision farming. These patterns demonstrate that, while the program has achieved significant progress, addressing these challenges through enhanced policy coordination and streamlined administrative processes can further empower rural women.

1. Introduction

The feminization of agriculture, characterized by the growing participation of women in the agricultural workforce, is a notable shift in rural economies of developing countries (Farnworth et al. 2022). This transition is primarily driven by male outmigration to urban areas and abroad for employment opportunities, leaving women to take on primary responsibilities for managing agricultural activities (Slavchevska et al. 2019). The recent literature suggests that the feminization of agriculture involves not only a redefinition of traditional gender roles but also complex socio-spatial challenges, as women often face unequal access to critical resources such as cultivable land, credit, technologies, and information. These shifts are shaped by persistent power dynamics, intersectional constraints, and the renegotiation of gender norms, which can simultaneously create opportunities for influence and reinforce existing inequalities (Leder 2022). These systemic inequities compromise women's ability to fully engage in and benefit from agricultural activities. Addressing these gaps requires targeted interventions that promote gender equality in ownership, access, and decision-making while fostering synergies across policy frameworks to enhance productivity and food security (Agarwal 2018)

Addressing gender-specific constraints in agriculture is essential to leverage women's contributions to agricultural productivity and food security. This requires equipping women farmers with equitable access to resources, alongside providing training and inclusive decision-making platforms (Doss 2018). Women's collectives, particularly in developing countries, have emerged as effective institutional mechanisms to address these gaps. These collectives enhance women's bargaining power, promote access to markets, foster community-based solutions, and encourage the adoption of sustainable agricultural practices (Agarwal 2020; World Bank 2020). By pooling resources, labor, and knowledge, women's groups have been shown to improve productivity, resilience, and sustainable growth, while simultaneously contributing to household food security (Raghunathan et al. 2019; Abdu et al. 2022).

Despite the documented empowerment benefits of women's collectives, significant research gaps persist in understanding their functional roles in agriculture, particularly in fostering innovation, improving productivity, and facilitating technology dissemination. Critics opine that women's collectives are sometimes viewed primarily as tools for achieving broader rural development goals rather than focusing on the holistic empowerment of the women involved, with financial and productivity outcomes prioritized over personal and social dimensions of women's empowerment (Guerrero et al. undated). The individual programs that involve women's collectives in agrarian or rural development are to be evaluated based on their ability to bring structural challenges within agricultural systems, such as enhancing farming system resilience and driving technological innovation, without compromising the women's empowerment and inclusion objectives. Against this backdrop, the present seeks to examine the challenges involved and contributions of women's collectives in the agricultural sector of Kerala, India.

Kerala presents a unique case, with the state mission on the women's collective "Kudumbashree" serving as a well-established network of women's collectives that have effectively combined economic and social empowerment with agricultural development (Agarwal 2020). It operates through a three-tier network: Neighbourhood Groups (NHGs), Area Development Societies (ADS), and Community Development Societies (CDS) (Praveen and Suresh 2015). Established in 1998, the mission now involves over 4.6 million women (in 2024), focusing on collective farming, microfinance, and credit access to promote economic self-sufficiency and enhance agricultural practices and community welfare (Kudumbashree 2024). Its decentralized yet organized institutional framework, spanning state to grassroots levels, facilitates economic participation among women, particularly those below the poverty line. By fostering cooperation and shared responsibility, Kudumbashree has demonstrated how collective action can mitigate structural barriers and promote sustainable agricultural development (Ramachandran and Das 2020).

The study evaluates the scope and challenges of the Kudumbashree mission for sustainable intensification of crop production, processing and marketing from the point of view of SHG participants. Insights from 52 Focus Group Discussions conducted across Malappuram and Trivandrum districts in Kerala reinforce the transformative role of collective action in empowering women and enhancing rural resilience. Despite challenges such as climate variability, inadequate infrastructure, and market instability, women's groups employ adaptive strategies, including sustainable farming practices, crop diversification, and the use of indigenous knowledge. These strategies not only stabilize livelihoods but also strengthen agricultural resilience and food security amidst growing pressures.

2. Context

According to the recent estimates, 59 per cent of India's employed women work in agriculture (ILO 2024). Despite this high participation, many women globally continue to bear a disproportionate burden of poverty, largely due to societal and structural barriers, as indicated earlier. To address these inequalities, the Central and State governments of India have established and nurtured women's SHGs, which have demonstrated positive effects on women's economic, social, and political empowerment (Aggarwal et al. 2021). are small, voluntary associations of individuals, typically from similar socio-economic backgrounds, who come together to address common challenges and improve their living conditions. These groups pool resources, skills, and talents to tackle socio-economic issues collectively (Swain and Wallentin 2009). In India, many SHGs are linked with banks to provide micro-credit access. The Reserve Bank of India has issued guidelines allowing SHGs, whether registered or unregistered, to open savings bank accounts with commercial banks, regional rural banks, and cooperative banks, with simplified Know Your Customer (KYC) procedures to facilitate the process (RBI 2023). In the early 1990s, the National Bank for Agriculture and Rural Development (NABARD) initiated the Self-Help Group Bank Linkage Programme (SHG-BLP) to integrate microfinance into the formal banking sector. This initiative aimed to shield the economically disadvantaged from exploitative moneylenders by providing comprehensive financial services, including savings, credit, insurance, and payment facilities (NABARD undated).

One of India's largest women's empowerment initiatives, Kudumbashree, was launched by the Government of Kerala in 1998 as part of the State Poverty Eradication Mission. Its aim is to eradicate poverty in Kerala's rural and urban areas through community development schemes, operating under the leadership of local self-governments. Kudumbashree is linked to local government institutions, facilitating poverty alleviation through a holistic approach that integrates resources and collective action (Jose, 2015). This initiative not only provides institutional support but also enables women to engage in income-generating activities, such as farming, food processing, and value addition, exemplifying a commitment to gender equity and sustainable agriculture at the grassroots level.

Under Kudumbashree, Neighborhood Groups (NHGs) of 20-40 women from economically weaker families are formed. Five volunteers from each group coordinate the activities, holding weekly meetings to manage operations (Irshad & Juman, 2015). Within this structure, Kudumbashree implements a range of activities to promote poverty alleviation, women's empowerment, and sustainable development. The mission strengthens community-based organizations through training, capacity building, and management information systems while facilitating microfinance services to enhance financial inclusion. It supports local economic development by promoting micro-enterprises, collective farming, animal husbandry, and producer companies, alongside marketing initiatives to improve market access. Social development initiatives include programs for destitution eradication, tribal upliftment, and education for differently abled children through BUDS schools. Kudumbashree also addresses gender issues through the State Gender Resource Center, Snehitha support networks, and health and hygiene programs. Special projects like the Attappady Development Project and Kudumbashree's international consultancy efforts further amplify its impact. Additionally, the mission implements centrally sponsored programs such as the National Rural Livelihoods Mission (NRLM) and the Mahila Kisan Sashaktikaran Pariyojana (MKSP) to empower women and youth, while urban projects like the Pradhan Mantri Awas Yojana (PMAY) address housing and livelihood needs of the urban poor. Collectively, these initiatives demonstrate a holistic approach to fostering inclusive growth and improving land use practices.

Within each NHGs, Joint Liability Groups (JLGs) are formed within the initiative to help women farmers secure agricultural credit, which often enhances productivity compared to individual farms (Agarwal, 2018). This model demonstrates a sustainable intervention strategy that adapts to local geographic, demographic, social, and economic conditions (Kumaramkandath, 2017). Women farmers within Kudumbashree have adopted new technologies and cultivation practices to counteract the adverse effects of climate change. These shifts highlight the role of women in promoting sustainable agricultural practices and food security. Traditionally, agricultural labor in India was divided along gender lines, with men handling physically demanding tasks while women focused on agricultural activities within the home. However, collective farming initiatives led by women's groups offer an opportunity to redefine these roles within a more inclusive, cooperative framework. Although substantial research exists on women's collectives and rural development, less attention has been given to their role in ensuring sustainable agriculture and food security, as well as the socio-economic factors that shape them. This study captures the evolving dynamics within women's collectives in two districts in Kerala, examining the economic, socio-political, and cultural factors that contribute to agricultural sustainability in the region.

3. Methodology

3.1. Study Area

Located on India's southwestern coast, Kerala spans 38,863 sq. km and is characterized by diverse topography and ecosystems, including coastlands, wetlands, plains, and hills (Fox et al., 2017). Though rooted in agriculture, the state now faces labour shortages, marketing inefficiencies, and climate uncertainties. For this study, Malappuram and Thiruvananthapuram districts in Kerala were purposively selected, each offering unique attributes for analysis. Malappuram, where Kudumbashree was first implemented, is predominantly rural and culturally diverse, providing an ideal location for examining Kudumbashree's grassroots impact (Irshad & Juman, 2015). Thiruvananthapuram, the state capital, represents an urban-rural nexus with numerous initiatives, offering insights into the broader impact of women's collectives in urban settings.

3.2. Sampling strategy

To investigate the role of women's collectives in sustainable agriculture and food security, 52 Focus Group Discussions (FGDs) were conducted with *Kudumbashree* groups in Thiruvananthapuram and Malappuram districts. Groups engaged solely in microfinance or non-agricultural enterprises (e.g., textile units) were excluded. The sample list of SHGs was generated through consultations with key informants, including:

- *Kudumbashree* State and District Mission personnel,
- *Kudumbashree* Chairpersons,
- Community Development Society (CDS) representatives, and
- SHG members.

The selected SHG units represented diverse performance levels and activity types. FGDs were guided by a semi-structured questionnaire designed to analyze both successful and unsuccessful groups. Discussions focused on factors influencing performance, constraints faced, and strategies for upgrading, marketing, and climate adaptation in agriculture and allied enterprises. In-depth interviews with select members clarified leadership roles and other emerging aspects. Ethical considerations, including confidentiality and voluntary participation, were emphasized, with an

ethical statement read before each session. In Thiruvananthapuram, 23 FGDs were conducted. Groups engaged in collective farming (e.g., floriculture and vegetable production), food processing (e.g., *Nutrimix* production), and catering services, including community kitchens. In Malappuram, 29 FGDs were conducted. Groups engaged in collective farming (e.g., paddy and fruit cultivation), food processing (e.g., spice mix production), and catering services. Each FGD involved 4 to 10 participants and lasted 1.5 to 2 hours. Summary notes from each discussion were compiled into detailed reports for analysis.

3.3. Data analysis

Each FGD summary was organized into individual documents, categorized by geographic location, SHG type, and class classification. This structure facilitated systematic import into NVivo, where each FGD document represented a unique case. Attributes like SHG type and class classification enabled segmentation for comparative analysis.

An initial coding structure was developed aligned with the research questions. Open coding identified recurring patterns and concepts across FGDs, later organized into a hierarchical framework of broader themes and sub-themes. NVivo's tools allowed for consistent categorization of text, ensuring uniformity in the coding process. Key themes, such as economic empowerment, social cohesion, and challenges faced by SHGs, emerged from the coding. NVivo's Query function was used to explore these themes, with matrix coding to cross-tabulate themes with case classifications (e.g., class and location). Cross-case analyses examined the impact of economic status, location, and SHG type on identified themes, revealing insights into thematic differences across groups.

NVivo's visualization tools, including diagrams and charts, illustrated relationships between themes and attributes, offering a structured overview of findings. These visualizations supported the interpretation of data by showing how specific attributes influenced thematic outcomes. The thematic analysis results, enriched with NVivo-generated visuals, were compiled into a comprehensive report discussing each theme in detail, emphasizing how class classification shaped participant experiences and thematic outcomes.

4. Results

4.1. Impact of climate variability on production and operations

The challenges posed by climate variability significantly affect production and operations within Kudumbashree collectives engaged in agriculture, microenterprises, and catering services. Seasonal weather fluctuations, including extreme heat, unpredictable rainfall, and humidity, introduce persistent risks across these sectors, impacting crop health, storage conditions, and work environments. The effects of erratic weather patterns have been particularly severe in collective farming. Water scarcity during the dry season and waterlogging during monsoon rains disrupt crop cycles, reducing yields and undermining productivity. For example, in marigold cultivation, members noted that "severe water scarcity during the summer has caused wilting and reduced flowering, while persistent rains have resulted in significant yield loss" (FGD-02), which directly affects the quality and quantity of harvest. Similarly, in early-stage millet cultivation, "unpredictable rains and sudden waterlogging spoiled the seedlings" (FGD-29), leading to substantial losses even before the crops

matured. Watermelon cultivation was challenging for a group “during summer, due to the high temperatures, the fruit was left partially dry” (FGD-05).

Humidity and temperature extremes present another formidable challenge, particularly in storage-dependent enterprises. High moisture levels during the monsoon period accelerate spoilage of stored grains and other products, reducing their shelf life and market value. A participant engaged in grain storage shared, “During the rains, stored grains turn mouldy within days and drying the grains become difficult. We lose both product and profit” (FGD-24). This situation is exacerbated by limited access to appropriate storage facilities, compelling members to store grains in temporary setups that are ill-equipped to handle extreme humidity. Temperature extremes impact operations in labour-intensive sectors, notably in catering and community kitchen services. High summer temperatures make working conditions challenging, especially in makeshift kitchen setups without adequate ventilation. In one instance, a community kitchen worker remarked, “Cooking under an asbestos sheet in peak summer is unbearable; the heat feels like it’s doubled inside” (FGD-24). Another participant added, “Monsoon winds blow away our makeshift roof, and we’re left with no protection while cooking” (FGD-47). These conditions not only affect worker comfort and safety but also pose food safety risks, as high temperatures can expedite spoilage and contamination during meal preparation.

In addition to these direct impacts, pest and disease outbreaks are exacerbated by changing climate conditions. Participants reported increased infestations during the rainy season, with one member involved in vegetable cultivation stating, “The rain brings pests that eat away at our crops and also results in rotting” (FGD-18). This climate-induced vulnerability to pests adds to the financial burden on SHG members, who must invest additional resources in pest control measures while facing unpredictable yields. While trying to keep their produce chemical-free, the infestation makes it difficult for them to maintain so.

The compounded impact of these climate-related challenges illustrates the vulnerability of Kudumbashree collectives to environmental variability. From water scarcity and pest attacks in agriculture to temperature constraints in catering services, each segment of SHG activity is directly affected. A participant summarized the broader concern, saying, “With every change in the season, we face a new set of challenges. It’s like the climate itself is working against us. The heavy rains make it difficult for us during transportation and also hinders the drying processes” (FGD-23). These insights underscore the critical need for adaptive strategies and support systems that can mitigate climate-related risks, improve resilience, and sustain productivity across sectors.

4.2. Market access and financial instability

Market access and financial instability present major challenges for Kudumbashree collectives involved in agriculture, microenterprises, and catering services. The absence of stable markets, fluctuating demand, unreliable pricing, and delayed payments create a cycle of uncertainty that hampers consistent income and sustainable growth for SHG members.

One prominent issue is the inconsistency of Kudumbashree markets, which are often limited to seasonal or occasional events, leaving members with few reliable outlets to sell their products. A member from a home-based enterprise explained, “The Kudumbashree markets only happen on

certain days, like festivals, and that's the only time we see good sales. Otherwise, we don't have a place to sell" (FGD-19). This seasonal market structure restricts continuous income, forcing members to depend on irregular sales periods that limit their ability to sustain operations year-round. In some cases, there is difficulty in sales as the market is a temporary setting, which exposes the food products to faster spoilage.

Fluctuating prices add to the financial challenges, especially in agricultural sectors. Many members expressed frustration with the lack of price stability, stating that "prices go up and down all the time; one season we earn well, and the next, the prices drop, and we are left with losses" (FGD-23). This unpredictability in pricing not only impacts income but also discourages investment in production as members hesitate to risk capital when future returns are uncertain.

Another significant issue is the delay in receiving payments from government programs and intermediaries in the value chain. A farmer shared, "When we sell to Supplyco, the payments take months to come. By then, we have no money left to reinvest in our farms" (FGD-25). Such delays in cash flow disrupt the operational cycle, especially for small-scale producers who rely on timely payments to cover immediate expenses and prepare for the next planting season.

The lack of a Minimum Support Price (MSP) further exposes SHG members to market volatility, as they are left at the mercy of prevailing market prices. Without MSP guarantees, members face increased financial vulnerability, as one participant explained: "If we had an MSP, we'd know there's at least a base price we'll get. But right now, we have to accept whatever price they offer" (FGD-05). Another farmer highlighted, "Despite investing significantly in cultivating pesticide-free products, we struggle to secure fair prices. Shopkeepers often reject our produce, opting for cheaper, pesticide-laden alternatives, instead" (FGD-10). The absence of an MSP undermines the financial stability of smallholder farmers, particularly in low-demand periods. The high commission of the available marketing platforms have limited the scopes of marketing as stated by a respondent, "We used to sell at Kudumbashree Home Shops but had to discontinue as they demanded a high commission of 40%."

Transportation difficulties exacerbate these market challenges, restricting the reach of Kudumbashree products to broader markets. As one member highlighted, "Transport is a big issue for us. We don't have the means to carry products to far markets, and local markets don't always have enough buyers" (FGD-05). High transportation costs limited the profits made as pointed out by a farmer, "Transportation costs are high as we need to transport our produce 30 km away for sale, where we are forced to sell at minimal prices" (FGD-32). These logistical limitations prevent SHG members from accessing more lucrative markets outside their local areas, thereby narrowing their customer base and reducing potential income.

Home-based enterprises face additional barriers due to limited networking opportunities and access to assured markets. One member commented, "We don't have connections to sell our products outside the Kudumbashree circles, so it's hard to find steady buyers and competition from bigger microenterprises is a hurdle we need to surpass" (FGD-09). This lack of market linkage limits the growth potential of these enterprises, which often struggle to move beyond small, localized sales.

The challenges related to market access and financial instability highlight the economic vulnerability of Kudumbashree collectives. The inconsistency of market access, coupled with price fluctuations,

payment delays, and logistical constraints, creates a precarious environment for SHG members. As one participant put it, “We work hard to produce quality products, but without reliable markets and fair prices, it’s a constant struggle just to keep going” (FGD-10). Addressing these barriers is essential for supporting the economic sustainability and growth of Kudumbashree enterprises, ensuring that members can build resilient livelihoods and continue contributing to local economies.

4.3. Operational constraints and technological limitations

The Kudumbashree collectives encounter numerous operational constraints and technological limitations, particularly in accessing and applying advanced agricultural practices. These constraints are largely shaped by regulatory restrictions, land tenure issues, and limited infrastructure, which hinder productivity and innovation within SHG activities.

One primary limitation arises from regulatory restrictions on using advanced technologies, such as drones for agricultural purposes. While many SHG members have received training in drone operations for precision agriculture, local regulations restrict drone usage in certain areas. A participant explained, “We trained for drone use to apply fertilizers more accurately, but authorities won’t allow us to fly them because of nearby electric lines and police stations” (FGD-02). These restrictions prevent SHG members from fully implementing the technology they have been trained to use, limiting their ability to improve efficiency in collective farming activities.

Land tenure issues also restrict technological advancements, particularly for those cultivating on leased lands. Landowners often hesitate to allow long-term investments in technology, fearing potential damage or encumbrances on their property. One member mentioned, “Unfortunately, we are not allowed to install a solar fence as the land is leased and the owner thinks it’ll affect their property” (FGD-32). This lack of control over leased land curtails SHG members' ability to invest in necessary infrastructure, such as irrigation systems or soil enhancement technologies, impeding the adoption of practices that could improve yields and resilience.

In addition to these structural constraints, inconsistent information dissemination from officials also disrupts operations within the Kudumbashree network. Communication between the Community Development Society (CDS) and Area Development Society (ADS) is often fragmented, leading to delays in crucial information reaching the SHG members. A participant noted, “By the time we get updates from the officials, it’s sometimes too late to act. We miss out on subsidies or important meetings” (FGD-11). These communication gaps slow down the decision-making process and create missed opportunities for members to benefit from timely resources and training.

Infrastructure limitations are another significant operational constraint, particularly in catering services and other microenterprises. A catering service participant described the challenge of working in poorly equipped facilities, saying, “Our kitchen setup is basic, just a small room with limited space for large orders. When there’s a rush, we can barely manage” (FGD-47). These constraints on space and facilities affect productivity and limit the potential for scaling operations, forcing members to operate within a constrained environment that limits growth and profitability.

Labor shortages further complicate operational efficiency, particularly in labour-intensive farming activities. With a limited workforce, SHG members must often take on additional roles, which impacts their productivity and leads to longer work hours. One farmer shared, “Sometimes it’s just a

few of us working the whole field. We can't afford to hire more help, so we have to do everything ourselves" (FGD-25). This shortage of labour puts additional stress on members and impacts the quality of output, as tasks are rushed or left incomplete due to time constraints.

Together, these operational constraints and technological limitations underscore the challenges Kudumbashree collectives face in improving productivity and innovation. From regulatory restrictions and land tenure issues to infrastructural and financial limitations, these obstacles create an environment where SHG members struggle to modernize and scale their operations effectively. As one participant summarized, "We have the skills and the drive to do more, but we're held back by so many hurdles. If these were removed, we could achieve much more" (FGD-19). Addressing these operational and technological barriers is crucial to unlocking the full potential of Kudumbashree initiatives, enabling members to improve efficiency, productivity, and overall resilience.

4.4. Strategies and innovations in climate resilience and market adaptation

In response to the significant challenges posed by climate variability and market instability, Kudumbashree members have developed a range of strategies and innovations to improve resilience and adapt to changing conditions. These adaptive practices, encompassing climate-resilient agricultural techniques, market diversification, and skill development, highlight the resourcefulness of SHG members as they seek sustainable solutions to safeguard their livelihoods.

One prominent climate resilience strategy involves adopting water-saving irrigation techniques, particularly drip irrigation, which allows for precise water management and conservation. As a member involved in watermelon cultivation explained, "We installed drip irrigation because it saves water, and we can still manage during the summer when water is scarce" (FGD-05). This method has proven effective in areas where water scarcity is an annual concern, enabling crops to survive dry seasons while minimizing water usage.

In collective farming, crop rotation and diversification are essential strategies for adapting to seasonal climate variations and protecting against pest infestations. A participant from a floriculture group noted, "We rotate between marigolds and vegetables according to the season to reduce pest problems and improve the soil" (FGD-02). Such practices not only enhance soil health but also reduce reliance on chemical pest control methods, contributing to more sustainable farming practices within the Kudumbashree network.

Integrated farming systems have also been adopted to enhance resilience by diversifying income sources. In one case, a member shared her experience with integrating dairy farming into crop cultivation: "We use cow dung and urine as fertilizers, and the leftover straw from the harvest goes to feed the cows. It's a full cycle that supports both our crops and livestock" (FGD-25). This approach not only improves soil fertility but also provides a secondary income from dairy products, reducing the reliance on a single crop and spreading economic risk.

To expand market reach and overcome local demand limitations, some members have explored digital platforms for selling their products. For instance, a herbal products unit in Thiruvananthapuram successfully introduced their products online, allowing them to reach customers beyond local markets. A participant explained, "Going online has helped us find new buyers who we couldn't reach through regular markets. Now, we have orders from nearby towns as

well” (FGD-43). By leveraging digital platforms, these SHGs have accessed broader markets and achieved more consistent sales, even during off-peak periods.

In the face of labour shortages, Kudumbashree members have invested in skill development, allowing them to perform tasks that traditionally required hiring additional workers. A participant shared her experience after receiving training: “Hiring labour, particularly for spraying pesticides, is challenging since most workers are men who are often occupied in other fields. However, with training programs now being offered by Krishi Bhavan, we have also started taking up pesticide spraying.” (FGD-44). By equipping themselves with new skills, members reduce dependency on external labor, empowering them to manage their operations independently and keep costs down.

In response to market instability, some SHG members have emphasized the need to upgrade product offerings and branding to appeal to diverse customer bases. One member from a paddy cultivation group mentioned plans to add spice mixes to their product line: “If we make products like spice mixes, it’ll help us expand beyond the paddy market and reach more customers. We just need a dryer and some branding support from CDS” (FGD-49). This diversification helps to mitigate risks associated with single-product reliance and improves the group’s ability to adapt to changing market demands.

Collective actions and initiatives, such as forming ‘Janakeeya Hotels’ (community kitchens), have been suggested as future strategies to enhance market stability and create new income avenues. A catering service member stated, “I want to start a community kitchen with other Kudumbashree members and deliver meals ourselves. If we get support from the Panchayat for an auto, we can even handle deliveries” (FGD-07). This model not only addresses local food needs but also provides steady income and enables the SHG members to connect directly with consumers, reducing reliance on middlemen.

Kudumbashree members employ a wide range of adaptive strategies to address the dual pressures of climate variability and market fluctuations. Through innovative approaches such as water-saving irrigation, crop rotation, digital marketing, skill development, and product diversification, SHG members demonstrate resilience and resourcefulness in their efforts to sustain and grow their enterprises. As one member put it, “We face many challenges, but with the right strategies, we can adapt and still move forward” (FGD-19). These strategies illustrate the proactive steps taken by SHG members to ensure both environmental sustainability and economic viability, showcasing the vital role of collective innovation in overcoming systemic challenges.

5. Interactions between institutions: How to make Kudumbashree units function better?

The institutional challenges faced by Kudumbashree women significantly affect their functioning and, in turn, their ability to ensure food security. Many groups struggle with delayed and insufficient subsidies, as one respondent noted, “Subsidies are insufficient, amounting to just Rs. 5000 per 2 lakhs borrowed, adding to our financial burden” (FGD-53). Similarly, the lack of timely access to adequate financial resources was emphasized by members engaged in collective vegetable farming,

where one respondent highlighted, “We received a loan of Rs. 2 lakhs, but the subsidy was only Rs. 5000, which barely helps in covering costs” (FGD-03). This financial strain is compounded by bureaucratic hurdles; another respondent explained, “Major constraint is the lack of ongoing government support. Initiating a program is not sufficient; continuous follow-ups and support from the government are necessary” (FGD-23). Bureaucratic delays in implementing programs were also noted as a bottleneck in marigold cultivation, where members depend heavily on the ward member's assistance for subsidies and support, raising concerns about sustainability if leadership changes (FGD-02).

Government programs such as the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA) play a dual role in SHG functioning. On the one hand, they provide much-needed employment and support for initial land preparation, as highlighted by a vegetable farming group: “Land preparation and clearing of weeds are done by the MGNREGA workers, enabling us to focus on cultivation” (FGD-03). The wages from MGNREGA also contribute to household income, indirectly supporting SHG members' participation in agricultural activities. On the other hand, the overlap in timing between MGNREGA activities and SHG responsibilities often creates scheduling conflicts, reducing the availability of members for group tasks. For instance, one group mentioned that despite their interest in expanding activities, members prioritized MGNREGA work due to its immediate income benefits, leading to decreased group engagement (FGD-03).

Political interference in Kudumbashree leadership further complicates matters, as highlighted by the observation that “leadership positions are often appointed from the ruling party's affiliates to align fund management with party interests. This leads to a lack of representation for apolitical voices in decision-making” (FGD-14). Favoritism was also reported by a collective farming group, where respondents noted that “SHGs with connections to political leaders get advanced information about schemes and benefits, putting others at a disadvantage” (FGD-03). This undermines the effectiveness of their functioning, hindering their potential to serve as a truly community-driven initiative for development. Furthermore, groups reported losing opportunities due to biases in resource allocation, with some ward members prioritizing groups aligned with their political affiliations over others (FGD-05).

Finally, gender norms significantly shape the participation and performance of women in SHGs, often acting as both barriers and motivators. Social expectations regarding women's roles in household management frequently conflict with their responsibilities within SHGs, limiting their availability for group activities. For instance, one respondent noted, “Managing both work and home has been smooth only because my family supports my participation in Kudumbashree; otherwise, balancing responsibilities would have been impossible” (FGD-02). In contrast, societal disapproval can discourage women from taking leadership roles or participating in training programs. A respondent recounted being criticized for attending a ten-day training session in Chennai, with others questioning her decision to leave her family behind, but she chose to ignore the remarks, finding strength in her family's support (FGD-02). Despite these challenges, SHGs often act as transformative spaces where women overcome restrictive gender norms. Members reported gaining confidence and independence, with one respondent stating, “Previously, I was afraid of handling banking tasks, but now I can independently go to banks and markets without seeking family approval” (FGD-01).

However, gendered biases persist, particularly in economic activities. Some women noted that selling produce in markets often resulted in lower earnings due to perceptions that female sellers were less competent, forcing them to rely on male relatives for marketing (FGD-06). Leadership dynamics within SHGs also reflect traditional gender roles, as some members are reluctant to assume leadership positions, citing household duties and societal resistance as barriers (FGD-04). On the positive side, participation in SHGs often challenges these norms, with many women reporting a shift in how they are perceived within their families and communities. As one respondent proudly noted, "Earlier, my opinions were not valued at home, but now my husband consults me before making financial decisions" (FGD-04). This dual dynamic of resistance and empowerment underscores the complex interplay of gender norms in shaping the functioning and success of SHGs.

Enhancing the role of Self-Help Groups (SHGs) in Kerala's agrarian development requires targeted interventions in mechanization, marketing, technology adoption, and land availability. Panchayat-level Custom Hiring Centers (CHCs) can be established with mobile services and digital booking platforms to facilitate access to agricultural machinery. Initiatives like women-focused equipment rental centers and collective technology investments can empower SHGs to modernize agricultural practices. To ensure sustainability, training and outreach through Krishi Bhavans can enhance skills, while integrating mechanized farming with MGNREGA can provide certified operators and align with rural development goals.

Marketing strategies must focus on creating a unified Kudumbashree brand with robust quality standards and certifications, complemented by regular inspections for continuous improvement. Expanding market access through e-commerce platforms and physical distribution channels, including partnerships with retail chains, can enhance product visibility and sales. Participation in trade fairs and leveraging digital marketing, coupled with the development of region-specific products, can further strengthen Kudumbashree's market presence. These efforts can not only increase income for women collectives but also create employment opportunities in production and marketing beyond Kerala.

Promoting technology adoption is another critical area. Establishing farmer schools can facilitate training in advanced practices like precision farming and climate-smart technologies, while cooperative farming models can pool resources to mitigate climate risks. Simplifying subsidy processes and linking crop insurance to technology adoption can reduce financial barriers and enhance resilience. Collective adoption of technologies by SHG members can lower costs and amplify benefits, fostering a collaborative approach to agricultural modernization.

Addressing land availability issues is essential for SHG-led agricultural expansion. Implementing long-term lease agreements under local governments can ensure fair and transparent access to land, while tailored subsidies and financial support can empower women collectives to utilize small and fragmented plots efficiently. Tax incentives for landowners and flexible lease renewal mechanisms can create sustainable partnerships between landowners and SHGs. By integrating these interventions with policy and institutional support, Kerala can harness the potential of SHGs to drive sustainable agrarian development and promote gender-inclusive economic growth.

6. Conclusion

Kudumbashree exemplifies the transformative potential of women's collectives in driving agrarian development through its robust organizational structure and grassroots adaptability. By fostering knowledge sharing, skill development, and collective innovation, it enables marginalized women to overcome barriers and actively contribute to sustainable agricultural practices and rural economies. This dual capability of structured top-level governance and localized flexibility empowers collectives to tailor solutions to their unique challenges, enhancing productivity, resilience, and inclusivity.

The findings of this study highlight the significant contributions of Kudumbashree collectives in addressing Kerala's agrarian challenges. Members have adopted climate-resilient practices such as crop rotation, drip irrigation, and integrated farming systems to mitigate the impacts of climate variability. Through collective marketing efforts and diversification, they have expanded access to markets and improved income stability. However, operational barriers such as limited access to advanced technologies, infrastructural constraints, and fragmented landholding systems persist, restricting their full potential. The study also underscores the role of community cohesion and leadership within collectives in navigating these challenges and sustaining growth.

Research on community-driven initiatives highlights how collective efforts foster skill development, reduce inefficiencies, and generate knowledge spillovers, creating sustained economic and social benefits. Kudumbashree's model integrates these dynamics, demonstrating the power of collective action to sustain growth and community well-being. However, persistent challenges such as market access limitations, climate variability, and infrastructural deficits underscore the need for targeted policy interventions and collaborative frameworks. Kudumbashree can further amplify its impact by addressing the abovementioned barriers, thereby offering a replicable model for inclusive, resilient, and sustainable rural development. Its success not only enhances the livelihoods of its members but also contributes to broader community transformation, showcasing the potential of grassroots initiatives to drive long-term economic and social progress.

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Chapter 6

Working with SHGs: The Morocco Story

[Paper title: Rural Collectives for Agriculture and Women Empowerment: Insights from Khemisset Province, Morocco]

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Abstract

Women's self-help groups (SHGs) have emerged as pivotal platforms for addressing the socio-economic and environmental challenges faced by rural women in the Global South. This study examines the role of SHGs in enhancing women's agency, fostering sustainable agricultural practices, and promoting resilience in Khemisset Province, Morocco. Based on data collected through Focus Group Discussions (FGDs) among 30 self-help groups (SHGs), the study highlights the unique role that these collectives play in integrating traditional practices, such as couscous processing and medicinal plant production, with modern processing techniques to capture niche markets. It reveals how localized leadership within SHGs drives collective action and sustains operations despite limited financial, administrative, and infrastructural support. Furthermore, SHGs build resilience to climate variability by promoting crop diversification and low-input, high-value production systems. Despite their success, systemic barriers, including restricted access to formal credit and entrenched gender norms, persist. These findings underscore the need for targeted policy support and capacity-building to amplify the transformative potential of SHGs.

1. Introduction

The role of women's self-help groups (SHGs) in fostering rural development and empowering women has gained increasing recognition globally, particularly in regions marked by socio-economic and environmental vulnerabilities (Basak and Chowdhury 2023). Women's participation in agriculture is indispensable to the socio-economic fabric of the Middle East and North Africa (MENA) region, particularly in Morocco (Baada and Najjar 2020). Rural women in Morocco play a crucial role in agricultural productivity, community cohesion, and food security. However, their contributions are often undervalued, constrained by cultural norms, limited access to resources, and economic vulnerabilities (Bossenbroek and Ftouhi 2021; Wafae and Sara 2021). Women's collectives have the potential to emerge as powerful mechanisms to bridge these gaps, fostering collective action for economic empowerment, social inclusion, and climate-resilient agriculture.

Agriculture accounts for approximately 14% of Morocco's GDP and employs 40% of its active population (ITA 2024). The sector is characterized by rainfed farming systems, making it highly vulnerable to climate variability (Abdelmajid et al. 2021). Women farmers are disproportionately affected by these vulnerabilities due to their reliance on small-scale, subsistence agriculture and their exclusion from resource ownership and technological access. SHGs address these challenges by promoting collective resource pooling, shared knowledge, and access to external support mechanisms. This Moroccan context offers a compelling case for studying SHGs, given the country's

reliance on agriculture and its significant gendered workforce. Women constitute 51% of the agricultural labor force but remain disproportionately excluded from land ownership, decision-making processes, and institutional support systems (FAO n.d.). The establishment of SHGs aligns with Morocco's broader development strategies, such as the Green Morocco Plan and the Green Generation Strategy, which emphasize gender equity and sustainability in agricultural development. These collectives serve as platforms for rural women to access financial services, market opportunities, and training in sustainable practices.

The socio-economic context of rural Morocco reveals a pressing need for inclusive models of agricultural development. Persistent droughts, rising input costs, and market volatility exacerbate the precarity of rural livelihoods (Ou-Salah et al. 2023). SHGs provide a structured approach to mitigate these risks, enabling women to diversify income sources, adopt climate-smart practices, and strengthen their bargaining power in markets. Against this backdrop, this paper examines the potential of SHGs in Morocco to advance sustainable agriculture, women's empowerment, and community resilience. Based on 30 Focus Group Discussions (FGDs) conducted across the Khemisset province in 2024, the study explores the dynamics of SHG formation, their operational challenges, and their impact on individual and collective capacities. The findings aim to contribute to the growing body of evidence on women's collectivization in agriculture, with implications for policy, institutional design, and grassroots mobilization.

This study is part of the Learning in Tandem (LiT) project, a collaboration between CIMMYT, IWMI, and Australian National University, aimed at examining the role of women's collectives in fostering climate-resilient agri-food systems in MENA and South Asia. In Morocco, the project focuses on understanding how SHGs enhance women's agency, facilitate sustainable practices, and integrate marginalized groups into the agricultural economy. The specific objectives of this study are to:

- Analyze the socio-economic and cultural factors influencing SHG formation and functioning.
- Assess the role of SHGs in promoting sustainable agricultural practices and enhancing women's economic participation.
- Identify challenges and opportunities in SHG operations, with a focus on institutional and policy frameworks.

By addressing these objectives, the study seeks to provide actionable insights for policymakers and development practitioners to support women's collectives as drivers of rural transformation.

2. Context

The feminization of agriculture in Morocco reflects a significant socio-economic shift, driven by male outmigration to urban areas and the persistent vulnerabilities of rural livelihoods. Women now constitute nearly half of the agricultural workforce, taking on a wide range of responsibilities, including crop production, livestock care, and food processing. However, their contributions often remain undervalued and concentrated in low-productivity, unpaid roles such as weeding, harvesting, and managing subsistence farms. This phenomenon has positioned women as the *de facto* managers of smallholder farms, presenting both opportunities for empowerment and challenges rooted in systemic inequalities.

While increased participation has expanded women's involvement in decision-making and farm management, it has also exacerbated their workload and exposure to risks. Women farmers in Morocco face limited access to land ownership, agricultural inputs, and financial services, further constraining their productivity and income potential. These challenges are compounded by traditional gender norms that hinder their access to training and institutional support. Despite these barriers, women's SHGs stand as a promising platform for addressing the needs of female farmers. By fostering collective action and resource pooling, SHGs enable women to overcome structural constraints, access markets, and adopt sustainable farming practices, thereby contributing to the resilience of rural communities.

The emergence of women's SHGs in Morocco represents a grassroots response to the structural and socio-economic challenges faced by rural women, particularly in the context of agricultural feminization. These groups have been catalyzed by various initiatives and organizations, such as the Green Morocco Plan (PMV) and the Green Generation Strategy, which emphasize gender integration in rural development. Local entities like the Khémisset Social Work Association and regional agricultural extension offices have played a pivotal role in motivating women to form SHGs and cooperatives. For instance, several groups in the Khemisset province were formed in response to drought conditions, leveraging institutional support to organize income-generating activities such as couscous production and honey processing. SHGs like Ait Belarbi and Dar Lemlih have successfully combined traditional practices with modern innovations, creating economic value while preserving cultural heritage. By fostering access to training, markets, and financial services, SHGs bridge the gap between marginalized women and institutional support systems. These collectives have become critical agents of economic empowerment, social inclusion, and climate resilience, addressing not only immediate income needs but also the long-term sustainability of rural livelihoods.

SHGs operate within a dynamic socio-economic and institutional framework that presents both significant challenges and opportunities. A primary challenge is the limited financial and infrastructural support available to SHGs. Many rural women lack access to credit and financial services, which constrains their ability to expand operations or invest in high-value economic activities. This issue is compounded by inadequate infrastructure, such as a lack of processing facilities and market access, which hampers the groups' ability to scale their initiatives. Additionally, cultural norms and traditional practices often restrict women's participation in decision-making and limit their opportunities to engage in leadership roles within cooperatives. Despite these constraints, SHGs in Morocco benefit from an enabling policy environment and support from local and international institutions. The Green Morocco Plan and the Green Generation Strategy prioritize women's empowerment in agricultural development, creating avenues for SHGs to access training, resources, and markets. Partnerships with organizations such as agricultural extension agencies and NGOs have facilitated capacity-building initiatives, enabling women to adopt sustainable farming practices and diversify their income sources. These support mechanisms underscore the potential of SHGs to not only address economic vulnerabilities but also contribute to broader goals of sustainable agriculture, climate resilience, and rural development.

3. Methodology

Study Region. Located in the Rabat-Sale-Kenitra region of Morocco, Khemisset Province is characterized by its diverse geography, including high-altitude plateaus (950–1300m), sandy plateaus, Zaer plateau, and arid zones. The region's agricultural landscape is predominantly rainfed, with only 5% of its Useful Agricultural Area under irrigation. Key agricultural activities include cereal and legume cultivation, livestock rearing, olive production, and the cultivation of aromatic and medicinal plants. These practices are highly vulnerable to recurring climate challenges, such as droughts, which impact cropping patterns and farmer resilience. The province serves as a critical area for understanding the socio-economic and environmental dynamics influencing rural livelihoods in Morocco.

Khemisset has a population of 542,221, with a rural majority (51.84%). The agricultural sector is predominantly small-scale, with 60% of farms being less than 5 hectares. Cooperative activity in the region includes 270 registered cooperatives, of which about 20% are women-led. Women's cooperatives mainly focus on food processing, animal products, and aromatic plant production, reflecting their pivotal role in local agricultural economies despite systemic challenges such as limited access to resources and markets.

Sampling Methodology. The study selected 30 women's cooperatives in Khemisset Province for Focus Group Discussions (FGDs). Cooperatives were chosen based on two main criteria: (1) Having been active for at least three years, and (2) Engagement in activities such as food processing, animal products, or aromatic plant production. Each FGD included 6–10 active members from the same cooperative, ensuring a representative understanding of group dynamics, governance, and challenges. This qualitative data was supplemented by an in-depth household survey targeting 125 women and 125 male farmers across 20 villages. The survey captured insights into gendered participation, impacts, and perceptions of cooperative activities. Semi-structured interviews focused on group formation, operational activities, governance structures, and challenges faced by women's cooperatives.

Table 1. Women Collectives Included in the Focus Group Discussions in Morocco

SHG Id.	Major activity of the SHG	Number of members in the SHG	Year of starting the SHG	Number of meetings in last 12 months
1	Aromatic and medicinal plant cultivation	5	2022	3
2	Aromatic and medicinal plant cultivation	13	2017	1
3	Preparation of honey-amlou toasts	5	2020	1
4	Honey production and marketing	7	2022	1
5	Aromatic and medicinal plant cultivation	11	2023	2
6	Aromatic and medicinal plant cultivation	8	2014	1
7	Aromatic and medicinal plant cultivation	8	2015	1
8	Honey production and marketing	6	2017	2
9	Farm chicken production, cattle breeding	6	2023	1
10	Production of semolina and aromatic and medicinal plants	7	2017	1
11	Processing and marketing of couscous	20	2010	1
12	Processing and marketing of couscous	9	2017	1
13	Processing and marketing of couscous	8	2012	2
14	Farming of olive, carob, and wheat	9	2020	1
15	Processing and marketing of couscous, spice production	8	2015	1
16	Legume cultivation	10	2017	1
17	Legume cultivation	6	2017	1
18	Processing and marketing of couscous	7	2014	1
19	Processing and marketing of couscous	5	2014	1
20	Honey production and marketing	9	2016	1
21	Aromatic and medicinal plant cultivation	11	2019	1
22	Processing and marketing of couscous	5	2022	1
23	Ecological tourism	7	2022	2
24	Spice production and marketing	11	2019	1
25	Fattening calves	5	2023	1
26	Carpet weaving and embroidery	8	2010	0
27	Honey production and marketing	5	2024	1
28	Milk production and marketing	8	2023	2
29	Honey production and marketing	9	2020	1
30	Honey production, value addition, and marketing	6	2018	1

4. Findings

4.1 Group Formation and Motivation

The formation of women's SHGs in Morocco reflects a complex interplay of economic necessity, cultural practices, and institutional support. These collectives have emerged as grassroots mechanisms to address the economic and social marginalization of rural women. Drawing from focus group discussions, this section explores the motivations underlying the establishment of SHGs, the role of leadership, and the external influences that shaped their evolution. The most recurring and prominent themes are summarized below.

Economic Necessity as a Catalyst. Economic vulnerability was a pivotal factor driving the formation of women's SHGs in Morocco. Many SHGs were established to provide supplemental household income and counter the financial precarity faced by rural families. Economic necessity served as both a unifying motivation and a catalyst for collective action, reflecting the broader socioeconomic challenges of rural communities.

The importance of stabilizing household income was highlighted by an SHG whose goal was to "keep a regular income" [#SHG 10]. This reflects the economic precarity experienced by many rural women, who rely on irregular and seasonal sources of livelihood. Similarly, another SHG sought to create a "social and economic place" where women could engage in productive activities that ensured both income generation and social solidarity [#SHG 11]. In some cases, the formation of SHGs was closely tied to ensuring a minimum standard of living for women and their families. For instance, one group noted that its motivation was to "guarantee an income for women" while fostering collaboration and mutual support among members [#SHG 12]. This dual emphasis on income security and collective well-being underscores the role of SHGs as both economic and social safety nets.

The recurring theme of economic necessity among Moroccan SHGs highlights their function as mechanisms for financial resilience in rural areas. These collectives represent a grassroots response to the challenges of economic insecurity, providing women with opportunities to pool resources, stabilize income, and enhance their economic agency. By addressing immediate financial needs while fostering long-term economic empowerment, SHGs play a vital role in supporting rural livelihoods and contributing to sustainable development.

Promoting Local Know-How. SHGs in Morocco are more than economic entities; they are stewards of local heritage. By promoting traditional knowledge and regional resources, these groups contribute to the preservation of cultural identity while supporting rural livelihoods. The promotion of traditional skills and the valorization of regional resources were central to the formation of many women's SHGs in the country. These groups leveraged indigenous knowledge and practices to establish niche economic activities that combined income generation with cultural preservation. By focusing on locally specific products and methods, SHGs not only created economic opportunities but also contributed to sustaining rural heritage. Several SHGs explicitly highlighted their commitment to preserving and promoting traditional knowledge. For example, one group focused on the "promotion of local know-how in the production of couscous," a staple food deeply embedded in Moroccan culture [#SHG 15]. Another group sought to enhance the "valorization of the region's legumes, particularly Zaer lentils," which hold economic and cultural significance in their locality.

[#SHG 16]. This approach underscores the role of SHGs in utilizing regionally distinctive products as a foundation for economic development.

The emphasis on local resources was further evident in an SHG that described its goal as the "valorization of Zaer legumes" while addressing the need for income-generating activities [#SHG 17]. By creating economic value from local crops, these groups tapped into existing skills and traditions to drive their operations. Such efforts also align with broader agricultural development strategies that focus on regional specialization and resource-efficient production. Moreover, some SHGs explicitly tied their economic activities to the creation of "social and economic places," where traditional practices could be integrated with modern economic systems to benefit their communities [#SHG 11]. This dual focus on economic resilience and cultural preservation reflects the strategic vision of these groups to balance tradition with market demands.

Leadership and Vision. Leadership emerged as a critical factor in the formation and success of women's SHGs in Morocco. Many groups were catalyzed by visionary leaders who provided strategic guidance, operational knowledge, and the motivation to mobilize resources and members. The role of leadership extended beyond logistical coordination, shaping the long-term vision and sustainability of these groups. In several SHGs, presidents with prior experience in cooperatives or community work played an instrumental role in initiating collective action. For example, the president of SGH #24, who had a history of working with women in craft cooperatives, brought her expertise to establish and lead her SHG. Similarly, the president OF SGH #25, utilized her doctoral scholarship to launch a livestock-rearing initiative, demonstrating how individual vision and resources can inspire broader community participation. The FGD participants acknowledged that her strategic approach not only created a sustainable livelihood model but also encouraged women in her community to join the initiative, fostering collective economic empowerment. In a third case [#SHG 28], the president of an SHG returned to the rural community with her spouse to lead a cooperative initiative. Their experience and commitment to rural development helped establish a framework for collective production and marketing, addressing local economic challenges.

These examples illustrate that effective leadership is often the cornerstone of successful SHGs. Visionary leaders provide the necessary momentum to overcome initial hurdles, build trust, and align group objectives with broader community development goals. For policymakers and development practitioners, investing in leadership development and capacity-building programs can amplify the impact of SHGs, ensuring their resilience and long-term sustainability.

External Influences and Institutional Support. External influences and institutional support played a crucial role in shaping the formation and operations of many women's SHGs in Morocco. The involvement of local associations was a recurring theme among SHGs. For instance, SHG #22 actively encouraged women to form a cooperative by leveraging their existing skills in craft production. This association not only motivated the women but also provided them with a framework to achieve financial independence. Such institutional encouragement highlights the importance of external actors in empowering marginalized groups to organize and mobilize. In some cases, environmental and economic pressures also led to the formation of SHGs with institutional backing. For example, after years of drought that impacted agricultural productivity, a group sought to address these challenges collectively with support from local agencies [#SHG 14]. This underscores the role of institutional support in helping communities adapt to adverse conditions while fostering collective

resilience. These examples demonstrate that external influences are often pivotal in overcoming initial barriers to group formation. Support from associations and institutions provides critical resources, including technical assistance, infrastructure, and market access, which are otherwise inaccessible to rural women. For development practitioners and policymakers, integrating institutional support mechanisms into rural development programs can enhance the effectiveness and sustainability of SHGs, enabling them to thrive in challenging environments.

4.2 Capacity & Activities

Capacity Development: Training and capacity-building efforts have been instrumental in the establishment and operational success of women's SHGs in Morocco. Many groups received targeted training to enhance their technical and managerial capabilities, enabling them to effectively manage activities and overcome challenges. This external support played a pivotal role in building the skills necessary for sustainable operations. Several SHGs explicitly highlighted the importance of training. For instance, one group noted that "training was received" to start their cooperative operations, which significantly boosted their confidence and preparedness [#SHG 1]. Similarly, another group emphasized that training provided them with foundational knowledge in areas critical to their activities [#SHG 2]. Some groups specifically acknowledged the relevance of capacity building, with members describing it as "important" for the smooth functioning of their collective enterprises [#SHG 4]. The training programs were diverse, covering topics such as technical production skills, cooperative management, and marketing strategies. For example, members of an SHG working on food processing received guidance on marketing and business planning, which enabled them to explore new market opportunities [#SHG 5]. Such targeted interventions not only improved productivity but also enhanced members' understanding of market dynamics and consumer preferences.

Economic activities. The economic activities undertaken by women's SHGs in Morocco reflect a blend of traditional practices and innovative approaches tailored to local contexts. These activities are vital for the economic sustainability of SHGs and serve as a platform for members to enhance their skills and improve their livelihoods. A majority of SHGs focused on regionally significant activities that utilized local resources and expertise. For example, one group began with the "extraction of essential and vegetable oils," eventually expanding to include soap and cosmetic product production [#SHG 1]. This progression underscores how SHGs evolve over time, diversifying their operations to meet market demands.

Food processing was a common activity among SHGs, with several groups engaging in couscous production. One cooperative initially prepared couscous but later shifted its focus to more profitable activities involving aromatic and medicinal plants [#SHG 2]. This highlights the adaptability of SHGs in identifying and capitalizing on high-value economic opportunities. Honey production and beekeeping also emerged as significant activities. One SHG specialized in "beehive products such as honey, pollen, and propolis," showcasing the economic potential of apiculture in rural settings [#SHG 3]. Another group focused exclusively on honey production, indicating a targeted approach to leveraging specific agricultural resources [#SHG 4].

While many SHGs demonstrated diversification and scaling up of activities, some were still in their early stages of development. For instance, one group noted that they had "just started" their operations, reflecting the nascent phase of their economic journey [#SHG 5]. This suggests that the scale and maturity of SHGs' economic activities vary widely, depending on factors such as available resources, market access, and institutional support. By combining traditional knowledge with innovative practices, these groups contribute to rural economic growth and resilience. Supporting SHGs in diversifying their activities and enhancing their market linkages can further amplify their impact on local economies.

Innovation and adaptation. SHGs in Morocco have demonstrated the ability to integrate traditional practices with modern innovations, adopt new techniques, and respond flexibly to changing market demands. These efforts not only enhance economic viability but also ensure the long-term relevance of their activities in dynamic rural economies. Several SHGs have embraced innovative approaches to improve productivity and expand their market reach. For example, groups involved in honey production have adopted modern beekeeping techniques, such as queen bee breeding and hive management, to increase output and product quality [FGD #3, FGD #4]. Similarly, SHGs producing couscous have integrated semi-mechanized processes to streamline production while retaining the artisanal qualities of their products [FGD #6]. These innovations reflect how SHGs balance efficiency with the preservation of traditional methods.

The blending of traditional knowledge with contemporary practices is a hallmark of SHG activities. Groups working with aromatic and medicinal plants, for instance, have leveraged their indigenous expertise in plant cultivation while adopting modern distillation techniques to produce essential oils [FGD #1, FGD #2]. This integration not only adds value to their products but also allows them to cater to niche markets that value both authenticity and quality. By combining old and new methods, SHGs enhance the marketability of their products while sustaining cultural heritage. Market dynamics often necessitate shifts in SHG activities, and many groups have shown remarkable flexibility in adapting to these changes. For instance, groups originally focused on producing staple foods like couscous have diversified into high-demand products such as herbal teas, cosmetic items, and organic honey [FGD #3, FGD #6]. These transitions are often driven by evolving consumer preferences and the need to remain competitive in local and international markets.

4.3. Goals and Challenges

Women SHGs in Morocco exhibit a strong focus on ensuring sustainability and pursuing long-term goals, despite facing numerous challenges in their formation and operational phases. Their approaches to sustainability are often rooted in diversification of activities, building financial independence, and strengthening group cohesion, which collectively contribute to their resilience and impact. A key aspect of sustainability for SHGs is the diversification of their economic activities. By expanding beyond traditional products, many groups have managed to reduce dependency on a single income stream and adapt to market demands. For instance, SHGs initially focused on couscous production have diversified into producing herbal teas, cosmetics, and organic honey, tapping into niche markets and catering to evolving consumer preferences [#FGD 6, #FGD 9]. This

strategy not only enhances financial stability but also opens up opportunities for scaling their operations.

Building financial independence has also emerged as a critical goal for many SHGs. Groups have sought to reduce reliance on external funding by developing internal resource pools and reinvesting profits into their activities. One SHG reported creating a shared savings fund, which enabled members to collectively finance the purchase of new equipment and raw materials [#FGD 4]. Such initiatives foster a sense of ownership and self-reliance, which are essential for long-term sustainability. Strengthening group cohesion is another fundamental aspect of ensuring the sustainability of SHGs. By fostering participatory decision-making and equitable benefit-sharing, many groups have been able to maintain high levels of member engagement and commitment. For example, one group emphasized the importance of involving all members in setting collective goals and distributing profits, which helped build trust and unity [#FGD 3]. This inclusive approach is critical for sustaining operations and preventing conflicts that could undermine the group's effectiveness.

SHGs also demonstrate forward-looking strategies by investing in capacity building and innovation. Groups have adopted modern technologies, such as semi-mechanized equipment for couscous production and distillation techniques for essential oils, to improve productivity and product quality [#FGD 2, #FGD 7]. They have also participated in training programs focused on business planning and digital marketing, enabling them to navigate competitive markets and expand their reach.

Despite these efforts, achieving long-term sustainability remains a work in progress for many SHGs. Challenges such as limited infrastructure, access to credit, and market volatility persist, requiring targeted interventions from policymakers and development organizations. These challenges underscored the difficulties of organizing and sustaining collective economic activities in rural contexts, particularly for groups with limited resources and institutional support.

Logistical Challenges. Logistical challenges emerged as a significant barrier for women's SHGs in Morocco, particularly during their formation and early operational stages. These challenges, often rooted in the lack of infrastructure and resources, significantly hindered the ability of SHGs to function effectively and achieve their objectives. A recurring issue faced by many SHGs was the absence of dedicated workspaces. One group reported working "six months without premises," highlighting how the lack of a permanent location impeded their ability to coordinate activities, store materials, and conduct regular meetings [#FGD 1]. This limitation not only affected the efficiency of operations but also impacted member participation, as some members found it challenging to engage consistently in the absence of a formal setup.

For many SHGs, the lack of essential equipment presented another critical barrier. One group cited difficulties in accessing tools necessary for the extraction and processing of essential oils, delaying their production timeline and reducing output quality [#FGD 3]. Similarly, SHGs involved in couscous production reported the absence of semi-mechanized equipment, which forced members to rely entirely on manual processes that were both time-consuming and labor-intensive [#FGD 6]. Storage and transportation constraints further complicated logistical operations for SHGs. Without proper storage facilities, several groups struggled to maintain the quality of perishable products such as honey and herbal teas. For example, one group reported losses due to the inability to store their honey production under optimal conditions [#FGD 4]. Transportation issues also posed significant

challenges, as groups often lacked the means to transport their products to markets, thereby limiting their ability to reach customers and expand their market base [#FGD 5].

In some cases, SHGs had to share infrastructure with other cooperatives or community organizations, leading to overcrowding and scheduling conflicts. For instance, one group mentioned that shared premises limited their access to equipment during peak production periods, creating delays and inefficiencies [#FGD 7]. This lack of dedicated resources underscores the competition for infrastructure in rural areas, which disproportionately affects women-led initiatives. Logistical challenges not only affected operational efficiency but also strained group dynamics. Inconsistent meeting spaces or delayed production timelines sometimes led to frustration among members, affecting morale and participation rates. For instance, one SHG reported that logistical constraints created tensions among members, particularly when resource allocation was perceived as unequal [#FGD 8].

To overcome these barriers, SHGs adopted various strategies. Internal resource pooling allowed some groups to temporarily rent spaces or purchase basic equipment. External support from government programs and NGOs also played a vital role in addressing infrastructure gaps. For example, one SHG received assistance from a local development agency to set up a small production facility, which significantly improved their operations [#FGD 9]. The logistical challenges faced by SHGs highlight the critical importance of infrastructure and resource support in fostering their growth. Addressing these barriers through targeted interventions such as providing dedicated workspaces, subsidized equipment, and storage and transportation facilities can enhance the operational efficiency of SHGs, enabling them to focus on scaling their activities and achieving their developmental goals.

Financial Constraints. Financial constraints emerged as a significant barrier, particularly during the initial phases of group formation. Many SHGs faced difficulties in securing funds for incorporation and operational expenses. For instance, one group highlighted challenges related to "incorporation costs and working capital," which posed considerable hurdles to starting their operations [#FGD 4]. The initial financial requirements often exceeded the capacity of group members, forcing them to pool limited personal resources or delay the launch of their activities. Another SHG struggled with "opening an account," citing reluctance from banks to engage with small cooperatives during their start-up phase [#FGD 5]. This lack of institutional support created a bottleneck, preventing these groups from accessing essential financial services such as loans or credit facilities.

The high cost of acquiring raw materials and production inputs added to the financial burden. For instance, one group reported difficulties in sourcing affordable supplies for their honey production, which affected their ability to meet demand and maintain profitability [#FGD 6]. Similarly, transportation costs for delivering products to distant markets often ate into their already thin profit margins. These systemic financial barriers were compounded by a lack of knowledge about alternative funding options, leaving many SHGs reliant on personal savings or irregular contributions from members. External shocks such as the COVID-19 pandemic further exacerbated these financial challenges. One SHG noted that their "start-up period coincided with COVID-19," leading to disruptions in financing and marketing efforts [#FGD 3]. The pandemic significantly reduced market access for SHGs, limiting their sales and depleting their already limited resources. For some groups,

this financial strain led to a reduction in member participation, as the economic uncertainty discouraged consistent involvement.

Despite these constraints, some SHGs found innovative ways to overcome financial barriers. Internal resource pooling was a common strategy, enabling groups to create a shared fund that could be used for small investments or emergency needs. Training programs also played a critical role, equipping members with financial literacy skills and knowledge of grant application processes [#FGD 2]. In certain cases, SHGs received external support from NGOs or government agencies, which provided seed funding or subsidies for specific activities. For instance, one group secured a grant to purchase essential equipment, reducing their reliance on external loans and improving their financial stability [#FGD 9].

Impact of External Shocks. The impact of external shocks posed significant challenges to women's SHGs in Morocco, particularly during their formation and early operational stages. These shocks, such as the COVID-19 pandemic and broader economic disruptions, exposed the vulnerabilities of SHGs and highlighted their need for adaptive strategies to ensure resilience. The pandemic was particularly disruptive, affecting SHGs across multiple dimensions. One group noted that their "start-up period coincided with COVID-19," which severely impacted their financing and marketing efforts [#FGD 3]. With movement restrictions and reduced market access, SHGs involved in the production of perishable goods, such as honey and herbal teas, faced significant difficulties in maintaining sales and income. Another SHG reported that the pandemic curtailed their ability to attend trade fairs and exhibitions, critical avenues for showcasing their products and expanding their customer base [#FGD 5]. The financial repercussions of the pandemic extended beyond lost revenue. Many SHGs struggled to maintain member participation during this period, as economic uncertainty discouraged women from committing to cooperative activities. One group highlighted that reduced member involvement during the pandemic strained their operations and slowed progress on key initiatives [#FGD 8]. This demonstrates how external shocks can disrupt both the financial stability and social cohesion of SHGs.

In addition to the pandemic, broader economic shocks, such as fluctuating input costs and market instability, further exacerbated the challenges faced by SHGs. For instance, one group reported that rising transportation costs made it increasingly difficult to deliver their products to markets, significantly reducing their profit margins [#FGD 6]. Similarly, fluctuations in the prices of raw materials, such as those needed for honey production, added another layer of financial strain [#FGD 4]. These economic shocks highlighted the precarious nature of SHGs' reliance on external markets and inputs.

Despite these challenges, some SHGs demonstrated resilience by adopting adaptive strategies. Diversification of activities emerged as a key approach, with some groups shifting their focus from staples like couscous to higher-demand products such as herbal teas and cosmetics during the pandemic [#FGD 9]. Others leveraged digital platforms to maintain market access, using social media and online marketplaces to reach customers when physical sales channels were unavailable [#FGD 7]. These strategies not only helped mitigate the immediate impacts of external shocks but also laid the foundation for more resilient operations in the future.

4.4 Early Impact of SHG on Women's Livelihoods

Through collective action, SHGs have provided women with access to income-generating avenues that were previously unavailable. For instance, one SHG reported that "three members left for a new job," demonstrating how participation in the group equipped them with skills and experience that opened pathways to formal employment [#SHG 2]. The economic benefits of SHGs extend beyond individual members, as increased household income often translates into improved living standards for entire families. By pooling resources, sharing knowledge, and building networks, SHGs empower women to overcome barriers to economic participation, enabling them to explore new markets and business opportunities. This role of SHGs as a steppingstone for employability underscores their critical contribution to advancing rural women's financial independence and stability.

The SHGs provide collective support mechanisms that enable women to address challenges such as family responsibilities and financial constraints. For instance, one SHG reported that members initially struggled with "family and financial constraints," which limited their ability to engage fully in cooperative activities. However, the group's supportive environment allowed them to balance personal responsibilities while actively contributing to the SHG's operations [#SHG 3]. By fostering a sense of solidarity and mutual aid, SHGs empower women to overcome cultural and structural limitations that traditionally restrict their economic participation. This collaborative approach not only alleviates individual challenges but also strengthens the group's cohesion, enabling members to collectively address broader socio-economic barriers within their communities.

The SHGs have become important platforms for fostering participatory governance and empowering members to take active roles in decision-making processes. By involving women in shaping the objectives and strategies of their cooperatives, SHGs promote inclusivity and ensure that all voices are heard. For example, one SHG emphasized that their approach aimed to "involve women in all stages of the process," ensuring that members had a voice in defining goals and activities [#SHG 6]. This participatory model not only enhances collective decision-making but also builds individual leadership skills and agency among members. Through regular meetings, transparent discussions, and shared responsibilities, SHGs create an environment where women can contribute meaningfully to the group's success, fostering a sense of ownership and accountability.

The organizational structure and activities of SHGs play a crucial role in building collective capacity, enabling groups to function effectively and address shared challenges. By working together, members pool their skills, knowledge, and resources, creating a strong foundation for sustained operations. SHGs often prioritize skill-sharing and peer learning, which enhance the technical and managerial competencies of the group as a whole. For instance, one SHG adopted a collaborative approach to tackle financial and logistical hurdles, using collective problem-solving to identify solutions and implement them efficiently [#SHG 3]. These practices not only improve the operational dynamics of SHGs but also strengthen group cohesion, creating a unified front to navigate external challenges. The ability to build and sustain collective capacity is a defining feature of successful SHGs, laying the groundwork for long-term impact.

Finally, the impact of SHGs on members and their communities varies depending on the stage of the group's development. Early-stage SHGs often experience limited changes as they focus on establishing basic operations and overcoming initial hurdles. For instance, one cooperative

described itself as being "in its embryonic stage," indicating that the full benefits of its activities had yet to materialize [#SHG 5]. In contrast, more mature SHGs report transformative outcomes, including increased member employability, stronger group cohesion, and expanded economic opportunities. This variation highlights the importance of sustained support and capacity-building initiatives tailored to the specific needs of SHGs at different stages. Providing targeted interventions during the early phases can accelerate progress, enabling SHGs to achieve their potential more effectively and deliver greater socio-economic impact over time.

5. Conclusion

This study highlights the transformative potential of women's SHGs in addressing systemic challenges faced by rural women in Morocco, particularly in Khemisset Province. A unique finding is the dual role of SHGs in bridging cultural and economic divides by leveraging traditional practices, such as Zaer legume valorization and artisanal couscous production, while integrating market-oriented innovations to cater to niche demands. Localized leadership within SHGs has also emerged as a pivotal factor, enabling groups to navigate entrenched gender norms and build cohesive operational frameworks despite minimal external support.

Another distinctive insight is the role of SHGs in climate resilience, where members adopt agroecological practices like water-efficient systems and the cultivation of drought-tolerant crops, mitigating vulnerabilities associated with Morocco's rainfed agriculture. However, systemic barriers persist, particularly in accessing formal credit and equitable resource allocation, which highlights gaps in the Green Morocco Plan's implementation at the grassroots level. This study uniquely underscores the importance of balancing external institutional support with internal capacity building, positioning SHGs as both economic engines and cultural custodians. By addressing these nuances, SHGs can be scaled as replicable models for fostering gender equity, cultural preservation, and sustainable rural development across the Global South.

Despite of the potentials, critical barriers that hinder the scalability of SHGs, including systemic reluctance from financial institutions to support small, women-led cooperatives and a dependence on external support for infrastructure and capacity building. These findings emphasize the need for interventions that go beyond generic policy support, focusing instead on creating robust financial frameworks and strengthening internal capacities for self-reliance. SHGs in Morocco are not just economic entities but adaptive systems addressing deeply rooted structural challenges, offering a replicable model for sustainable rural transformation in similar socio-economic contexts.

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Chapter 7

Comparing Self-Help Groups in Kerala and Morocco: Learning in Tandem

[Paper under Development]

Self-Help Groups (SHGs) have emerged globally as pivotal platforms for addressing socio-economic, environmental, and gender equity challenges in rural communities. This paper compares SHG initiatives in Kerala, India, and Khemisset Province, Morocco, highlighting their contributions to women's empowerment, sustainable agricultural practices, and resilience building. Drawing from the Learning in Tandem (LiT) project's focus on contextualizing women's collectives across MENA and South Asia, the comparison explores major topics, including cultural dynamics, institutional frameworks, economic activities, leadership structures, climate resilience strategies, and the role of external support. These themes provide insights into how SHGs adapt to unique socio-cultural contexts and institutional environments to foster inclusivity, strengthen collective action, and promote climate-resilient agricultural systems. By integrating qualitative and quantitative evidence, the analysis underscores the transformative potential of SHGs while addressing persistent socio-cultural and infrastructural constraints.

Formation and Structure

In India, SHGs are typically small, informal groups of 10-20 women from similar socio-economic backgrounds. They are often linked to formal financial institutions through programs like the SHG-Bank Linkage Program initiated by the National Bank for Agriculture and Rural Development (NABARD) in 1992. It is operated through a structured framework, heavily supported by the state government and linked to national programs. On the other hand, In MENA regions, including Morocco, Women's SHGs are generally organized by non-governmental organizations (NGOs) or international agencies. They may operate as cooperatives or community-based organizations, focusing on collective economic activities and social support. The main focus is on blending traditional practices with modern techniques to address market and climate challenges.

Role of SHGs in Women Empowerment

Both Kerala and Morocco leverage SHGs as platforms for enhancing women's empowerment. In Kerala, SHGs are integrated into the Kudumbashree program, a state-run initiative that combines economic and social empowerment with agricultural development. Kudumbashree promotes women's leadership, participation in decision-making, and access to resources, helping women overcome patriarchal barriers. Similarly, in Morocco, SHGs enable women to take leadership roles, fostering agency and independence in a traditionally male-dominated society. Moroccan SHGs serve as stewards of local heritage, preserving traditional practices while integrating women into economic activities like couscous production and aromatic plant cultivation.

Despite these shared goals, the nature of empowerment differs. Kerala's Kudumbashree benefits from a structured, top-down framework that connects SHGs to broader government initiatives. In contrast, Moroccan SHGs rely heavily on organic, grassroots leadership and external support from NGOs and international collaborations. This difference underscores the varying institutional environments shaping the SHGs.

Institutional and Policy Frameworks

The institutional support for SHGs varies significantly between Kerala and Morocco. In Kerala, the Kudumbashree initiative is deeply embedded in state governance, linking SHGs to government programs like the Mahatma Gandhi National Rural Employment Guarantee Act (MGNREGA). This integration ensures sustained financial and infrastructural support for SHGs. Conversely, Moroccan SHGs operate within a less centralized framework, relying on policies such as the Green Morocco Plan and support from local NGOs. While these initiatives emphasize gender equity and sustainability, the lack of strong government oversight often limits their scalability.

Both systems have their merits and challenges. Kerala's approach provides consistent support and resources but risks bureaucratic inefficiencies and delays. Morocco's reliance on local leadership and NGO involvement fosters adaptability but struggles with financial and infrastructural constraints.

Economic Activities and Market Access

Economic activities in SHGs reflect the socio-economic contexts of Kerala and Morocco. Kerala's SHGs emphasize collective farming, food processing, and micro-enterprises, supported by Kudumbashree's structured networks. These activities promote sustainable agricultural practices and local food security. Moroccan SHGs, on the other hand, focus on a blend of traditional and modern practices, such as couscous production, honey cultivation, and aromatic plant processing. These activities align with Morocco's emphasis on niche markets and cultural preservation.

Market access challenges are significant in both contexts. Kerala's SHGs face issues like irregular markets, pricing volatility, and high transportation costs. Similarly, Moroccan SHGs struggle with limited credit access, high input costs, and inadequate infrastructure. Both regions demonstrate the need for improved market linkages, e-commerce integration, and financial support to enhance SHG sustainability.

Climate Resilience Strategies

Climate resilience is a critical focus for SHGs in both Kerala and Morocco, given the vulnerability of rural livelihoods to environmental changes. In Kerala, SHGs adopt practices such as crop rotation, drip irrigation, and integrated farming systems to mitigate climate variability. Moroccan SHGs employ crop diversification and low-input, high-value production systems to adapt to droughts and market fluctuations. These strategies highlight the resourcefulness of SHGs in addressing environmental challenges through localized solutions.

Cultural Contexts and Gender Norms

The cultural contexts of Kerala and Morocco significantly shape the functioning of SHGs. In Kerala, patriarchal norms influence women's participation, though Kudumbashree provides a structured pathway for overcoming these barriers. In Morocco, entrenched gender norms limit women's access to resources and leadership opportunities. However, SHGs in Morocco play a dual role in addressing these norms while preserving cultural heritage through traditional practices.

Leadership and Community Dynamics

Leadership dynamics differ notably between Kerala and Morocco. In Kerala, leadership often emerges within the structured Kudumbashree framework, where positions are linked to government-appointed roles. In Morocco, SHG leadership is more organic, driven by individuals who mobilize resources and inspire collective action. This grassroots approach fosters a strong sense of ownership among members but may lack the institutional backing needed for long-term sustainability.

Conclusion

Kerala and Morocco present compelling cases of how SHGs can empower women, promote sustainable practices, and address rural challenges. While Kerala benefits from a robust institutional framework, Morocco demonstrates the power of grassroots leadership and cultural preservation. Both models underscore the importance of adapting SHG structures to local contexts and challenges. By learning from each other's experiences, SHGs in Kerala and Morocco can enhance their effectiveness. Kerala can adopt Morocco's emphasis on cultural integration, while Morocco can benefit from Kerala's structured institutional support. Together, these insights contribute to the broader discourse on leveraging SHGs for rural transformation and women's empowerment in diverse socio-economic landscapes.